R and clinical/lab data

2026 edition

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2026-04-17

Contents

Welcome **5**

1 Dataframes, vectors and numbers **7**

1.1 Dataframes 7

1.2 Import file 7

1.3 Vectors 9

1.4 Save a dataframe 20

1.5 Home exercise 20

1.6 Bonus: how patient data were created 22

2 Strings, Factors and Dataframe Handling **25**

2.1 What we will see today 25

2.2 Strings and Factors 25

2.3 Dataframe Handling 34

2.4 Save your results 49

2.5 Home exercise 50

2.6 Bonus: how the clinical data were created 51

3 Data visualization **53**

3.1 Data visualization in R with ggplot2 53

3.2 Geom 54

3.3 Labs 67

3.4 Scales 68

3.5 Facet 74

3.6 Theme 76
3.7 Save the plots 78
3.8 Advanced examples 78

Welcome

Welcome to this course, here you can find the notebooks used in the PhD course of 2026.

Chapter 1

Dataframes, vectors and numbers

1.1 Dataframes

The usual data you will deal with are tabular (rows x columns) data, usually improperly called "Excel files".

The standard formats for raw data files in data analyses are *.csv*, *.tsv* and *.txt* files; these should be our starting point.

1.2 Import file

The first thing to do is to import the file into R session. We can do it with several functions, but the standard one for *.csv* files is `read.csv`.

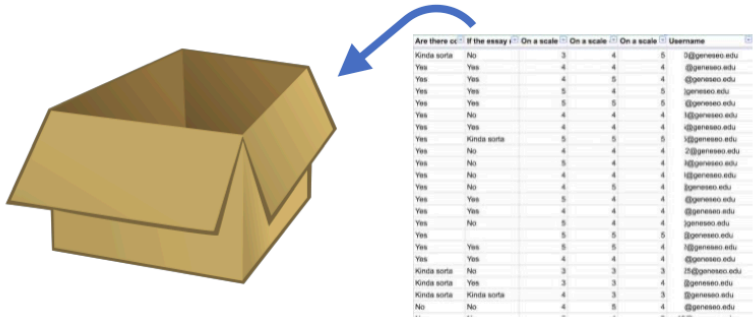
Let's see how to do so:

```
my_data <- read.csv("data/input_df.csv")
```

1.2.0.1 Notes about variables

Variables are the "memory" of an R session, they represent box where R stores data/values you want.

You can think them as boxes that have a name on them:



Are there co	If the essay i	On a scale	On a scale	On a scale	Username
Kinda sorta	No	3	4	5	3@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	Yes	4	5	4	6@geneseo.edu
Yes	Yes	5	4	5	)geneseo.edu
Yes	Yes	5	5	5	6@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	Yes	4	4	4	4@geneseo.edu
Yes	Kinda sorta	5	5	5	1@geneseo.edu
Yes	No	4	4	4	2@geneseo.edu
Yes	No	5	4	4	8@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	No	4	5	4	)geneseo.edu
Yes	Yes	5	4	4	@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	No	5	4	4	)geneseo.edu
Yes		5	5	5	)geneseo.edu
Yes	Yes	5	5	4	1@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Kinda sorta	No	3	3	3	75@geneseo.edu
Kinda sorta	Yes	3	3	4	)geneseo.edu
Kinda sorta	Kinda sorta	4	3	3	)geneseo.edu
No	No	4	5	4	@geneseo.edu

There are important **rules** about variables:

- In each variable you can store one single item (a number, a string, a dataframe, a list, ...)
- variable names can start only with a letter (uppercase or lowercase)
- variable names can contain **only** letters, numbers, underscores (_) and dots (.). **No spaces, no dash (-) or any other special characters**
- there could be only ONE variable with a specific name, saving an object with the same of another results in the substitution of the first one
- suggestion: variable names should be meaningful to you (and to the ones you share the code with): e.g. `x` is not a good name, while `patients_glucose_data` is preferred

1.2.1 First steps on a dataframe

When importing a dataframe, there is a fundamental step to do: *inspection*. Inspection helps us to understand how the dataframe is structured, which type

of data we have, which columns etc.

To do it, we use 3 functions that you need to remember (tatoo yourself): `summary`, `str` and `head`. Let's see them:

```
summary(my_data)
```

```
     sample        timepoint           genotype          treatment              area        inter
 Min.   :  1.00   Length:192         Length:192         Length:192         Min.   :3000   Min.
 1st Qu.: 48.75   Class :character   Class :character   Class :character   1st Qu.:3259   1st Qu.
 Median : 96.50   Mode  :character   Mode  :character   Mode  :character   Median :3517   Median
 Mean   : 96.50                                                           Mean   :3508   Mean
 3rd Qu.:144.25                                                           3rd Qu.:3769   3rd Qu.
 Max.   :192.00                                                           Max.   :3988   Max.
```

```
str(my_data)
```

```
'data.frame':   192 obs. of  6 variables:
 $ sample   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ timepoint: chr  "P1" "P1" "P1" "P1" ...
 $ genotype : chr  "wt" "wt" "wt" "wt" ...
 $ treatment: chr  "n" "n" "n" "n" ...
 $ area     : num  3125 3500 3773 3538 3936 ...
 $ intensity: num  16015465 19449815 21118210 18100075 20718554 ...
```

We can see how many rows and columns we have, as well as which columns, the data type of each and the first entries.

Lastly, `head` is the function that shows us the first 6 entries of the dataframe, as a table:

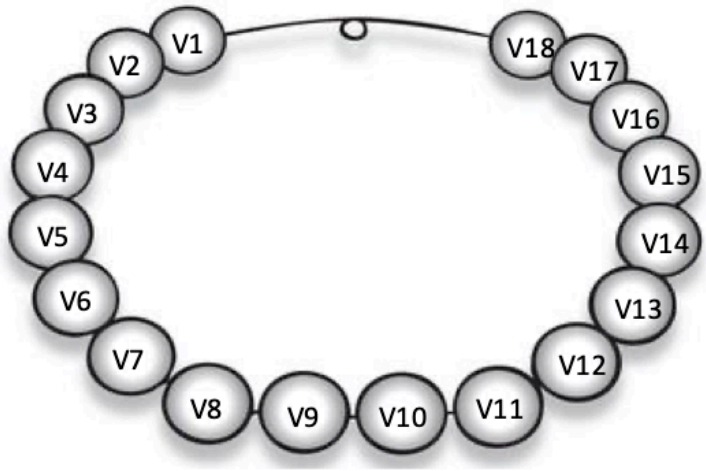
```
head(my_data)
```

```
  sample timepoint genotype treatment     area intensity
1      1        P1       wt         n 3124.709  16015465
2      2        P1       wt         n 3499.559  19449815
3      3        P1       wt         n 3773.000  21118210
4      4        P1       wt         n 3538.320  18100075
5      5        P1       wt         n 3936.014  20718554
6      6        P1       wt         n 3646.389  19507110
```

1.3 Vectors

Dataframes are organized and read by columns. Each column is a **vector** in R.

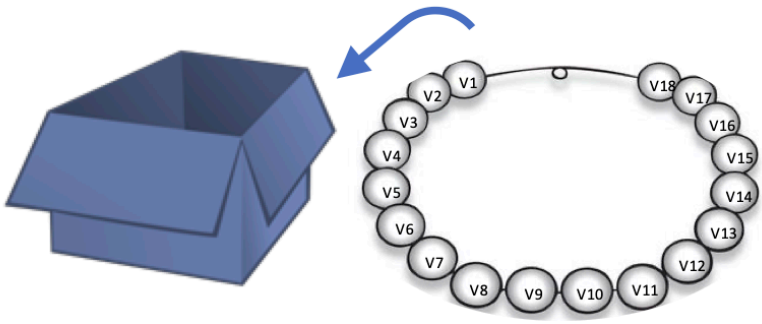
A vector is a collection of data of the **same type**; you can think of a vector as a pearl necklace, with each pearl being a single value:

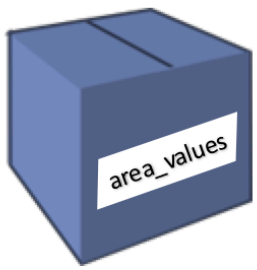


1.3.1 Accessing dataframe columns

For exercise purposes, we can create a new variable storing the values of the area column of the dataframe. To extract a single column from a dataframe, we use the structure `dataframe_name$column_name`.

```
area_values <- my_data$area
```





1.3.2 Accessing values of a vector: slicing

Slicing is the operation of extract values out of a variable (in this case a vector). It is done to answer these questions:

- *Which is the first value of the vector?*
- *Which is the value at position n of a vector?*
- *Which are the values from position n to m?*

There are different ways in which we can slice a vector, and today we will see indexing (through positions).

The syntax is:

- `variable_name[position]` to get one single value
- `variable_name[start:end]` to get elements from `start` to `end`

REMEMBER: In R, the first element is at index 1.

```
# extract one single value

first_area_value <- area_values[1]
print(first_area_value)
```

```
[1] 3124.709
```

```
# extract range of values
from_fifth_to_ninth_areas <- area_values[5:9]
print(from_fifth_to_ninth_areas)
```

```
[1] 3936.014 3646.389 3073.109 3717.063 3887.006
```

Q: *What 5:9 does in R?* A: It creates a vector containing numbers from 5 to 9, that's why it works:

```
print(5:9)
```

```
[1] 5 6 7 8 9
```

Q: *And to get last element?*

A: We need to know the lenght of the vector and use it as index. To get the length of a vector we use the function `length`:

```
length_area <- length(area_values)
print(length_area)
```

```
[1] 192
```

```
last_area <- area_values[length_area]
print(last_area)
```

```
[1] 3808.121
```

Exercise 1.1. Which is the 5th value of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

fifth_intensity <- intensity_values[5]

print(fifth_intensity)
```

```
[1] 20718554
```

Or in just one step:

```
fifth_intensity <- my_data$intensity[5]

print(fifth_intensity)
```

```
[1] 20718554
```

1.3.2.1 Replace a value by index

We can replace of a vector using indexes.

Note: new values should be of the same type (numbers for numbers, strings for strings ecc), otherwise everything will be changed to strings (most of the time).

To change a single value, we can type `variable_name[position] <- new_value`; to change multiple values `variable_name[start:end] <- new_values`.

1.3.3 Numbers in R

Now that we know how to get and extract numbers from numeric vectors, let's see what we can do with numbers in R.

1.3.3.1 Declare a new numeric variable

To declare a new numeric variable, we use the statement `variable_name <- number`:

```
conversion_factor <- 1000

print(conversion_factor)
```

```
[1] 1000
```

```r
print(typeof(conversion_factor))
```

```
[1] "double"
```

Decimal point numbers wants the dot `.` as decimal separator:

```r
decimal_n <- 215.6
print(decimal_n)
```

```
[1] 215.6
```

```r
print(typeof(decimal_n))
```

```
[1] "double"
```

1.3.3.2 Arithmetical operations

We can perform all sort of matematical operation:

```r
# sum

print(2 + 5)
```

```
[1] 7
```

```r
# subtraction

print(7 - 2.5)
```

```
[1] 4.5
```

```r
# multiplication
print(12 * 5)
```

```
[1] 60
```

```r
# division
print(105 / 43)
```

```
[1] 2.44186
```

```r
# exponential
print(5 ** 2)
```

```
[1] 25
```

```r
print(5 ^ 3)
```

```
[1] 125
```

```r
# square root
print(sqrt(12))
```

```
[1] 3.464102
```

We can combine them all to do more complex operations. For example, you can resolve this equation $x^2 - 7x + 12 = 0$. We know that the formula to resolve this equation is: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, so we can reconstruct them in R:

```r
a <- 1
b <- -7
c <- 12

x1 <- (-b + sqrt(b**2 - 4*a*c)) / (2 * a)
x2 <- (-b - sqrt(b**2 - 4*a*c)) / (2 * a)

print(x1)
```

```
[1] 4
```

```r
print(x2)
```

```
[1] 3
```

Exercise 1.2. Which are the last 3 values of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

length_intensity <- length(intensity_values)

third_to_last_index <- length_intensity - 2

last_3_intensities <- intensity_values[third_to_last_index:length_intensity]

print(last_3_intensities)
```

```
[1] 22965783 22027811 22338574
```

1.3.4 Vector-number operations

What we have seen so far on arithmetical operations between single numbers, can be applied also to numerical vector x single number operations.

In our data, area is expressed in µm^2 and we want to convert it into mm^2. To do so, we should divide each value of the area by a factor of 1,000,000; the cool thing about R is that this is done automatically when we use the statement `numeric_vector <operand> number`:

```
division_factor = 1e6

areas_mm2 <- area_values / division_factor

print(head(areas_mm2))
```

```
[1] 0.003124709 0.003499559 0.003773000 0.003538320 0.003936014 0.003646389
```

```
print(head(area_values))
```

```
[1] 3124.709 3499.559 3773.000 3538.320 3936.014 3646.389
```

This is true for ANY arithmetical operations:

```
print(head(intensity_values))
```

```
[1] 16015465 19449815 21118210 18100075 20718554 19507110
```

```
print(head(intensity_values + 10))
```

```
[1] 16015475 19449825 21118220 18100085 20718564 19507120
```

1.3.5 Operations between vectors

It is possible also to perform element-wise operations between vectors.

Let's load a new dataset with some patient data, and calculate the BMI of each patient.

```r
patient_df <- read.csv("data/patient_data.csv")

str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

BMI formula: $\frac{\text{weight in kg}}{(\text{height in m})^2}$

To do so, we should transform the height in m, and then perform the operation. We will save the results as a **new column** of the dataframe; this can be done with the syntax `dataframe_variable$new_column_name <- vector` or `dataframe_variable$["new_column_name"] <- vector` (we will see on day 2 the strings).

```r
# Get height in m
patient_df$height_in_m <- patient_df$height / 100

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m
1          1 75.82265 185.6147    1.856147
2          2 61.77137 212.2587    2.122587
3          3 91.65377 188.3604    1.883604
4          4 66.38117 176.2986    1.762986
5          5 80.42689 174.0509    1.740509
6          6 79.32015 173.7906    1.737906
```

```r
patient_df["BMI"] <- patient_df$weight / (patient_df$height_in_m ** 2)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI
1          1 75.82265 185.6147    1.856147 22.00768
2          2 61.77137 212.2587    2.122587 13.71059
3          3 91.65377 188.3604    1.883604 25.83279
4          4 66.38117 176.2986    1.762986 21.35733
5          5 80.42689 174.0509    1.740509 26.54902
6          6 79.32015 173.7906    1.737906 26.26220
```

1.3.6 Sumamry statistics of numeric vectors

Usually, when dealing with numeric data we want to have some summary statistics on a specific data (e.g. mean values, median, quartile, standard deviation, …).

In R there are many built-in functions that can help us in doing so:

- `mean` to calculate the mean
- `sd` to calculate the standard deviation
- `variance` to calculate the variance
- `median` to calculate the median
- `quantile` to calculate the nth quantile of a distribution
- `round` to round decimal values to n decimal places
- `sum` to sum all the values of the vector

- many others

They all have in common the syntax: `name_of_the_function(vector)`.

```
intensity_mean <- mean(intensity_values)
print(intensity_mean)
```

```
[1] 19419175
```

```
height_sd <- sd(patient_df$height)
print(height_sd)
```

```
[1] 26.73183
```

```
BMI_median <- median(patient_df$BMI)
print(BMI_median)
```

```
[1] 25.23683
```

```
first_quartile_weight <- quantile(patient_df$weight, probs = 0.25)
print(first_quartile_weight)
```

```
     25%
68.25118
```

```
rounded_BMI <- round(patient_df$BMI, 1)
print(head(rounded_BMI))
```

```
[1] 22.0 13.7 25.8 21.4 26.5 26.3
```

Exercise 1.3. Scale BMI values (formula: $scaled_i = \frac{x_i - \overline{x}}{sd(x)}$)

Solution

Here is a step-by-step solution:

```
# calculate BMI mean
mean_BMI <- mean(patient_df$BMI)

# calculate BMI sd
sd_BMI <- sd(patient_df$BMI)

# Calculate scaled BMIs
patient_df$scaled_BMI <- (patient_df$BMI - mean_BMI) / sd_BMI

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
```

That is exactly what the formula `scale` do underneath:

```r
# Calculate scaled BMIs
patient_df$scaled_BMI_function <- scale(patient_df$BMI)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI scaled_BMI_function
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706         -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320         -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959         -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319         -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031          0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341         -0.01120341
```

1.4 Save a dataframe

We are satisfied with this preliminary edit of the patients' data, so we can save the data to a dedicated file.

We use `write.csv` function:

```r
write.csv(patient_df, # what we want to save
          file = "output/patients_data_BMI.csv", # name of the file
          row.names = FALSE) # we'll see row names in day 2
```

1.5 Home exercise

For next time, if you want, you can try to do this exercise:

1. Starting from patient data (raw), load the file
2. Inspect it

3. You know that you have to give drug A to each patient so that the final concentration is 10 mg/kg (mg of drug every kg of patient). Calculate how much drug you should give to each patient.
4. Given that a single stock of drug A is 5 g, how many stocks you have to order?

```r
# 1. Load the file
patient_df <- read.csv("data/patient_data.csv")
```

```r
# 2. Inspect it
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

```r
str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
# 3. Calculate drug for each patient

patient_df$drugA_quantity_mg <- patient_df$weight * 10
head(patient_df)
```

```
  patient_id   weight   height drugA_quantity_mg
1          1 75.82265 185.6147          758.2265
2          2 61.77137 212.2587          617.7137
3          3 91.65377 188.3604          916.5377
4          4 66.38117 176.2986          663.8117
5          5 80.42689 174.0509          804.2689
6          6 79.32015 173.7906          793.2015
```

```
# 4. How many stocks you have to order?

total_drug_A_mg <- sum(patient_df$drugA_quantity_mg)

total_drug_A_g <- total_drug_A_mg / 1000

stock_weight <- 5

number_of_stocks <- total_drug_A_g/stock_weight

print(number_of_stocks)
```

```
[1] 15.03539
```

To be precise, we can use `ceiling` function that round to the upper integer:

```
number_of_stocks <- ceiling(number_of_stocks)

print(number_of_stocks)
```

```
[1] 16
```

1.6 Bonus: how patient data were created

```
set.seed(3564) # set randomness seed
l <- 100 # define number of patients
patient_data <- data.frame("patient_id" = 1:l, # create a vector from 1 to l
                           "weight" = rnorm(n = l, mean = 75, sd = 10), # create a vec
                           "height" = rnorm(n = l, mean = 175, sd = 25))

write.csv(patient_data,
          file = "data/patient_data.csv",
          row.names = F)
```

```r
head(patient_data)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

Chapter 2

Strings, Factors and Dataframe Handling

2.1 What we will see today

Today we will look at some patient data, and we will answer these questions:

- Are data consistent?
- Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?
- Which group has the highest mean creatinine? (2 different datasets)

2.2 Strings and Factors

Today we will work with a clinical dataset containing information about patients enrolled in a multi-site trial. Let's import it and, as always, **inspect it first**:

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------------------------------------------
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------------------------------
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

```r
clinical_data <- read.csv("data/clinical_data.csv")
```

```r
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : chr  "Milan" "Milan" "Turin " "Milan" ...
 $ visit_date      : chr  "2020-03-23" "2023-10-31" "2023-06-27" "2018-10-25" ...
 $ age             : int  66 62 54 64 51 55 71 63 61 55 ...
 $ sex             : chr  "Male" "Male" "Female" "Male" ...
 $ smoking         : chr  "FORMER" "current" "Curent" "Never" ...
 $ treatment       : chr  "Control" "Drug_B" "Drug_B" "Drug_A" ...
 $ stage           : chr  "I" "II" "III" "IV" ...
 $ weight_kg       : num  78.8 87.6 61.6 79.2 96.1 69 74.1 59.8 88.2 71 ...
 $ height_cm       : num  181 170 162 178 182 ...
 $ sbp_mmhg        : int  143 138 144 122 127 142 125 113 105 179 ...
 $ dbp_mmhg        : int  90 78 84 93 89 92 62 84 80 75 ...
 $ glucose_mgdl    : int  102 53 80 103 123 110 119 82 138 131 ...
 $ cholesterol_mgdl: int  250 207 180 129 178 173 239 212 173 204 ...
 $ crp_ngml        : num  1.37 11.09 0.81 1.63 1.25 ...
```

```r
summary(clinical_data)
```

```
  patient_id            site            visit_date             age            sex
 Length:240         Length:240         Length:240         Min.   :28.00   Length:240
 Class :character   Class :character   Class :character   1st Qu.:50.75   Class :charac
 Mode  :character   Mode  :character   Mode  :character   Median :58.00   Mode  :charac
                                                           Mean   :57.83
                                                           3rd Qu.:66.25
                                                           Max.   :97.00
   height_cm        sbp_mmhg        dbp_mmhg       glucose_mgdl    cholesterol_mgdl
 Min.   :147.4   Min.   : 89.0   Min.   : 52.00   Min.   : 43.00   Min.   : 88.0    Mi
 1st Qu.:162.8   1st Qu.:117.0   1st Qu.: 75.00   1st Qu.: 89.75   1st Qu.:177.0    1s
 Median :168.5   Median :130.5   Median : 82.00   Median :104.00   Median :197.0    Me
 Mean   :169.3   Mean   :131.4   Mean   : 82.02   Mean   :104.43   Mean   :197.1    Me
 3rd Qu.:176.2   3rd Qu.:143.2   3rd Qu.: 89.25   3rd Qu.:119.00   3rd Qu.:219.5    3r
 Max.   :193.4   Max.   :195.0   Max.   :108.00   Max.   :181.00   Max.   :295.0    Ma
```

```r
head(clinical_data)
```

```
  patient_id   site visit_date age    sex      smoking treatment stage weight_kg height_cm sbp_mm
1     PT-001  Milan 2020-03-23  66   Male       FORMER   Control     I      78.8     181.4      1
2     PT-002  Milan 2023-10-31  62   Male      current    Drug_B    II      87.6     169.9      1
3     PT-003 Turin  2023-06-27  54 Female       Curent    Drug_B   III      61.6     162.1      1
4     PT-004  Milan 2018-10-25  64   Male        Never    Drug_A    IV      79.2     178.2      1
5     PT-005  Milan 2018-02-19  51   Male       FORMER    Drug_B     I      96.1     182.3      1
6     PT-006  Milan 2023-02-06  55 Female never smoker   Control    II      69.0     165.9      1
```

You will notice that several columns contain text: `patient_id`, `sex`, `treatment`, `stage`, `smoking`, `site`, `diagnosis_code`. In R, text data can be represented in two fundamentally different ways: as **strings** (also called *characters*) or as **factors**. Let's understand the difference.

A **string** (or character) is simply a sequence of text. R treats it as a raw piece of text and does not impose any structure on it.

A **factor** is a *categorical variable*: it has a fixed, predefined set of allowed values called **levels**. Under the hood, R stores factors as integers (one integer per level) and keeps a table that maps integers to labels. This makes factors memory-efficient and very useful for statistical modelling (e.g., in regression, factors are automatically handled as categorical predictors) and for plotting purposes.

When to use strings and when factors?

Use **string** when…	Use **factor** when…
Text is free-form or unpredictable	The variable takes a limited set of known values
You need to manipulate the text	Order matters (e.g., stage I < II < III < IV)
Unique identifiers (patient IDs)	Input to statistical models

As some of our columns fall in the conditions for factors, let's convert them with the function `factor`:

```r
clinical_data$sex       <- factor(clinical_data$sex)
clinical_data$treatment <- factor(clinical_data$treatment)
clinical_data$site      <- factor(clinical_data$site)
clinical_data$smoking   <- factor(clinical_data$smoking)
```

```r
# For stage, the categories have a natural ORDER: we declare an ordered factor
clinical_data$stage <- factor(clinical_data$stage,
                              levels = c("I", "II", "III", "IV"),
                              ordered = TRUE)

str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 1 4 1 1 1 1 1 1 1 ...
 $ visit_date      : chr  "2020-03-23" "2023-10-31" "2023-06-27" "2018-10-25" ...
 $ age             : int  66 62 54 64 51 55 71 63 61 55 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 1 2 2 ...
 $ smoking         : Factor w/ 7 levels "Curent","current",..: 4 2 1 6 4 7 3 6 2 1 ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 1 3 3 2 3 1 1 1 2 1 ...
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 1 2 3 4 1 2 1 3 3 2 ...
 $ weight_kg       : num  78.8 87.6 61.6 79.2 96.1 69 74.1 59.8 88.2 71 ...
 $ height_cm       : num  181 170 162 178 182 ...
 $ sbp_mmhg        : int  143 138 144 122 127 142 125 113 105 179 ...
 $ dbp_mmhg        : int  90 78 84 93 89 92 62 84 80 75 ...
 $ glucose_mgdl    : int  102 53 80 103 123 110 119 82 138 131 ...
 $ cholesterol_mgdl: int  250 207 180 129 178 173 239 212 173 204 ...
 $ crp_ngml        : num  1.37 11.09 0.81 1.63 1.25 ...
```

Now `summary` gives us much more useful information on those columns:

```r
summary(clinical_data)
```

```
  patient_id            site      visit_date             age           sex
 Length:240         Milan :109   Length:240         Min.   :28.00   Female:118   Curent
 Class :character   Rome  : 61   Class :character   1st Qu.:50.75   Male  :122   curren
 Mode  :character   Turin : 33   Mode  :character   Median :58.00                forme
                    Turin : 37                      Mean   :57.83                FORME
                                                    3rd Qu.:66.25                never
                                                    Max.   :97.00                Never
                                                                                 never

    dbp_mmhg       glucose_mgdl    cholesterol_mgdl    crp_ngml
 Min.   : 52.00   Min.   : 43.00   Min.   : 88.0    Min.   : 0.260
 1st Qu.: 75.00   1st Qu.: 89.75   1st Qu.:177.0    1st Qu.: 2.140
 Median : 82.00   Median :104.00   Median :197.0    Median : 3.600
 Mean   : 82.02   Mean   :104.43   Mean   :197.1    Mean   : 4.673
 3rd Qu.: 89.25   3rd Qu.:119.00   3rd Qu.:219.5    3rd Qu.: 5.742
 Max.   :108.00   Max.   :181.00   Max.   :295.0    Max.   :28.150
```

Notice how `summary` now shows the count per level for factors (e.g. how many Male vs Female), while before it just said `character`.

Moreover, we can see that in smoking column, we have different labels with the same meaning (never, Never, never smoker for never). R (as well as any programming language) is case sensitive, so it treats those 3 as different labels, even if for us they mean the same thing.

IMPORTANT: labels should be consistent across datasets when you fill them.

As we want to edit those to be consistent, we need to change the column back to be a character, with `as.character` function:

```
clinical_data$smoking   <- as.character(clinical_data$smoking)
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 1 4 1 1 1 1 1 1 1 ...
 $ visit_date      : chr  "2020-03-23" "2023-10-31" "2023-06-27" "2018-10-25" ...
 $ age             : int  66 62 54 64 51 55 71 63 61 55 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 1 2 2 ...
 $ smoking         : chr  "FORMER" "current" "Curent" "Never" ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 1 3 3 2 3 1 1 1 2 1 ...
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 1 2 3 4 1 2 1 3 3 2 ...
 $ weight_kg       : num  78.8 87.6 61.6 79.2 96.1 69 74.1 59.8 88.2 71 ...
 $ height_cm       : num  181 170 162 178 182 ...
 $ sbp_mmhg        : int  143 138 144 122 127 142 125 113 105 179 ...
 $ dbp_mmhg        : int  90 78 84 93 89 92 62 84 80 75 ...
 $ glucose_mgdl    : int  102 53 80 103 123 110 119 82 138 131 ...
 $ cholesterol_mgdl: int  250 207 180 129 178 173 239 212 173 204 ...
 $ crp_ngml        : num  1.37 11.09 0.81 1.63 1.25 ...
```

We can now proceed.

2.2.1 Operations on strings

The main package for string operations in R (and the one we will use throughout the course) is `stringr`, which is part of the `tidyverse`. All its functions start with `str_` to make them easy to remember.

2.2.1.1 Character substitution

Let's start by looking at how to change characters in a string: from "Curent" to "current" using `str_replace(vector, old_pattern, new_pattern)`.

```
print(unique(clinical_data$smoking))
```

```
[1] "FORMER"       "current"      "Curent"       "Never"        "never smoker" "former
```

```
clinical_data$smoking <- str_replace(clinical_data$smoking, "Curent", "current")
print(unique(clinical_data$smoking))
```

```
[1] "FORMER"       "current"      "Never"        "never smoker" "former"       "never"
```

Then we can unify "FORMER" and "former" by transforming the former to lowercase with `str_to_lower`:

```
print(unique(clinical_data$smoking))
```

```
[1] "FORMER"       "current"      "Never"        "never smoker" "former"       "never"
```

```
clinical_data$smoking <- str_to_lower(clinical_data$smoking)
print(unique(clinical_data$smoking))
```

```
[1] "former"       "current"      "never"        "never smoker"
```

We have converted both FORMER to former and Never to never. We lastly have to change "never smoker" to "smoker".

Exercise 2.1. Change "never smoker" to "smoker" and make smoker as factor.

Solution

```
clinical_data$smoking <- str_replace(clinical_data$smoking, "never smoker", "never")
clinical_data$smoking <- factor(clinical_data$smoking, levels = c("never", "former", "c

summary(clinical_data)
```

```
  patient_id            site      visit_date            age           sex         smo
 Length:240         Milan :109   Length:240         Min.   :28.00   Female:118   never
 Class :character   Rome  : 61   Class :character   1st Qu.:50.75   Male  :122   forme
 Mode  :character   Turin : 33   Mode  :character   Median :58.00                curre
                    Turin : 37                      Mean   :57.83
                                                    3rd Qu.:66.25
                                                    Max.   :97.00
    dbp_mmhg       glucose_mgdl    cholesterol_mgdl    crp_ngml
 Min.   : 52.00   Min.   : 43.00   Min.   : 88.0    Min.   : 0.260
```

```
 1st Qu.: 75.00   1st Qu.: 89.75   1st Qu.:177.0    1st Qu.: 2.140
 Median : 82.00   Median :104.00   Median :197.0    Median : 3.600
 Mean   : 82.02   Mean   :104.43   Mean   :197.1    Mean   : 4.673
 3rd Qu.: 89.25   3rd Qu.:119.00   3rd Qu.:219.5    3rd Qu.: 5.742
 Max.   :108.00   Max.   :181.00   Max.   :295.0    Max.   :28.150
```

For sake of completeness, we could have done all of this with a single function `replace_values`:

```
clinical_data$smoking <- replace_values(clinical_data$smoking, c("never smoker", "Never") ~ "neve
                                        c("Curent") ~ "current",
                                        c("FORMER") ~ "former")
```

Q: *Why there are 2 Turin?* A: Let's check in details the possible labels of that column with function `levels`:

```
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

Here's the hitch, some have a trailing space after Turin. We can get rid of them using `str_trim` function:

```
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

```
clinical_data$site <- factor(str_trim(clinical_data$site))

print(levels(clinical_data$site))
```

```
[1] "Milan" "Rome"  "Turin"
```

```
print(summary(clinical_data))
```

```
  patient_id           site      visit_date             age           sex         smoking      t
 Length:240         Milan:109   Length:240         Min.   :28.00   Female:118   never  :129   Cor
 Class :character   Rome : 61   Class :character   1st Qu.:50.75   Male  :122   former : 57   Dru
 Mode  :character   Turin: 70   Mode  :character   Median :58.00                current: 54   Dru
                                                   Mean   :57.83
                                                   3rd Qu.:66.25
                                                   Max.   :97.00
```

```
    dbp_mmhg       glucose_mgdl    cholesterol_mgdl    crp_ngml
 Min.   : 52.00   Min.   : 43.00   Min.   : 88.0    Min.   : 0.260
 1st Qu.: 75.00   1st Qu.: 89.75   1st Qu.:177.0    1st Qu.: 2.140
 Median : 82.00   Median :104.00   Median :197.0    Median : 3.600
 Mean   : 82.02   Mean   :104.43   Mean   :197.1    Mean   : 4.673
 3rd Qu.: 89.25   3rd Qu.:119.00   3rd Qu.:219.5    3rd Qu.: 5.742
 Max.   :108.00   Max.   :181.00   Max.   :295.0    Max.   :28.150
```

2.2.1.2 Other useful functions to handle strings

A few more `stringr` functions you will find very useful:

Function	What it does
`str_detect(x, pattern)`	Returns TRUE/FALSE: does the string contain the pattern?
`str_starts(x, pattern)`	Does the string *start* with the pattern?
`str_ends(x, pattern)`	Does the string *end* with the pattern?
`str_pad(x, width, side, pad)`	Pads a string to a given width
`str_sub(x, start, end)`	Extracts a substring by position

2.2.2 Operations on multiple columns

2.2.2.1 Split a column and merge multiple columns

Sometimes a single column contains multiple pieces of information that could be split into separate columns. Look at `visit_date`:

```
head(clinical_data$visit_date)
```

```
[1] "2020-03-23" "2023-10-31" "2023-06-27" "2018-10-25" "2018-02-19" "2023-02-06"
```

We can see that the year of visit varies a lot, maybe we should take it into account. We can split this into 3 separate columns using `separate` function:

```
clinical_data <- separate(clinical_data,
                          col = visit_date,          # column to split
                          into = c("visit_year", "visit_month", "visit_day"), # new c
                          sep = "-",                 # separator
```

```
                              remove = FALSE)           # keep the original column

head(clinical_data)
```

```
  patient_id  site visit_date visit_year visit_month visit_day age    sex smoking treatment stage
1     PT-001 Milan 2020-03-23       2020          03        23  66   Male  former   Control     I
2     PT-002 Milan 2023-10-31       2023          10        31  62   Male current    Drug_B    II
3     PT-003 Turin 2023-06-27       2023          06        27  54 Female current    Drug_B   III
4     PT-004 Milan 2018-10-25       2018          10        25  64   Male   never    Drug_A    IV
5     PT-005 Milan 2018-02-19       2018          02        19  51   Male  former    Drug_B     I
6     PT-006 Milan 2023-02-06       2023          02        06  55 Female   never   Control    II
```

The opposite operation — merging multiple columns into one — is done with `unite`:

```
# example: unite back together
clinical_data <- unite(clinical_data,
                       col = "visit_year_month",          # name of the new column
                       visit_year, visit_month,            # columns to merge
                       sep = "-",                        # separator
                       remove = FALSE)                   # keep original columns

head(clinical_data)
```

```
  patient_id  site visit_date visit_year_month visit_year visit_month visit_day age    sex smokin
1     PT-001 Milan 2020-03-23          2020-03       2020          03        23  66   Male  forme
2     PT-002 Milan 2023-10-31          2023-10       2023          10        31  62   Male curren
3     PT-003 Turin 2023-06-27          2023-06       2023          06        27  54 Female curren
4     PT-004 Milan 2018-10-25          2018-10       2018          10        25  64   Male   neve
5     PT-005 Milan 2018-02-19          2018-02       2018          02        19  51   Male  forme
6     PT-006 Milan 2023-02-06          2023-02       2023          02        06  55 Female   neve
  cholesterol_mgdl crp_ngml
1              250     1.37
2              207    11.09
3              180     0.81
4              129     1.63
5              178     1.25
6              173     3.46
```

2.2.3 Column and row names

Every dataframe has **column names** (the variable names) and **row names** (by default a number, but we can change it).

To retrieve column names:

```
colnames(clinical_data)
```

```
 [1] "patient_id"       "site"             "visit_date"       "visit_year_month" "visit
[10] "smoking"          "treatment"        "stage"            "weight_kg"        "heigh
[19] "crp_ngml"
```

To rename specific columns, we can assign a new vector of names to `colnames`, or use the tidyverse function `rename` (we will see this with `dplyr` below):

```
clinical_data = rename(clinical_data, "diastolic_bp_mmhg" = "dbp_mmhg", "systolic_bp_mm

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"        "visit_year_month"  "
 [9] "sex"               "smoking"           "treatment"         "stage"             "
[17] "glucose_mgdl"      "cholesterol_mgdl"  "crp_ngml"
```

For row names:

```
# by default, row names are row numbers
head(rownames(clinical_data))
```

```
[1] "1" "2" "3" "4" "5" "6"
```

```
# we could set patient_id as row names
rownames(clinical_data) <- clinical_data$patient_id

head(rownames(clinical_data))
```

```
[1] "PT-001" "PT-002" "PT-003" "PT-004" "PT-005" "PT-006"
```

However, **be careful** with row names: they are not a proper column of the dataframe, so they don't follow along easily in many tidyverse operations. It is generally better to keep your identifiers as a regular column.

2.3 Dataframe Handling

The main verbs we will learn today are:

- `mutate` — add or modify columns
- `select` — choose which columns to keep
- `filter`/`filter_out` — keep/remove rows that satisfy a condition
- `group_by` + `summarize` / `reframe` — aggregate data by groups
- `left_join` / `inner_join` — merge two dataframes

2.3.1 Create new columns with mutate

`mutate` creates new columns or modifies existing ones.

A key feature of `dplyr` is the **pipe operator** `|>` (or `%>%` in older tidyverse style). It passes the result of one operation directly to the next, making code much more readable:

You can read `|>` as *"and then"*.

```
# Add a BMI column (we already know this formula!)
clinical_data <- clinical_data |> # take clinical_data and then
  mutate(height_m = height_cm / 100,
         BMI = weight_kg / height_m^2,
         BMI = round(BMI, 1))

head(clinical_data)
```

```
       patient_id  site visit_date visit_year_month visit_year visit_month visit_day age    sex s
PT-001     PT-001 Milan 2020-03-23          2020-03       2020          03        23  66   Male
PT-002     PT-002 Milan 2023-10-31          2023-10       2023          10        31  62   Male c
PT-003     PT-003 Turin 2023-06-27          2023-06       2023          06        27  54 Female c
PT-004     PT-004 Milan 2018-10-25          2018-10       2018          10        25  64   Male
PT-005     PT-005 Milan 2018-02-19          2018-02       2018          02        19  51   Male
PT-006     PT-006 Milan 2023-02-06          2023-02       2023          02        06  55 Female
       glucose_mgdl cholesterol_mgdl crp_ngml height_m  BMI
PT-001          102              250     1.37    1.814 23.9
PT-002           53              207    11.09    1.699 30.3
PT-003           80              180     0.81    1.621 23.4
PT-004          103              129     1.63    1.782 24.9
PT-005          123              178     1.25    1.823 28.9
PT-006          110              173     3.46    1.659 25.1
```

`mutate` can also create columns based on conditions, using `if_else` or `case_when`:

```
# Classify BMI into categories
clinical_data <- clinical_data |>
  mutate(BMI_class = case_when(
```

```
    BMI < 18.5 ~ "Underweight",
    BMI < 25   ~ "Normal",
    BMI < 30   ~ "Overweight",
    BMI >= 30  ~ "Obese"
  ),
  BMI_class = factor(BMI_class, labels = c("Underweight", "Normal", "Overweight", "Obes
  )

summary(clinical_data$BMI_class)
```

```
Underweight      Normal  Overweight       Obese
         94          42          91          13
```

2.3.2 select

`select` chooses which **columns** to keep (or drop). This is useful when working with wide dataframes where you only need a subset of variables.

```
# keep only demographic and clinical variables
demo_cols <- clinical_data |>
  select(patient_id, site, age, sex, treatment, stage, BMI, BMI_class)

head(demo_cols)
```

```
       patient_id  site age    sex treatment stage  BMI   BMI_class
PT-001     PT-001 Milan  66   Male   Control     I 23.9 Underweight
PT-002     PT-002 Milan  62   Male    Drug_B    II 30.3      Normal
PT-003     PT-003 Turin  54 Female    Drug_B   III 23.4 Underweight
PT-004     PT-004 Milan  64   Male    Drug_A    IV 24.9 Underweight
PT-005     PT-005 Milan  51   Male    Drug_B     I 28.9  Overweight
PT-006     PT-006 Milan  55 Female   Control    II 25.1  Overweight
```

To **drop** specific columns, use `-` before the column name:

```
colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"        "visit_year_month"  "
 [9] "sex"               "smoking"           "treatment"         "stage"             "
[17] "glucose_mgdl"      "cholesterol_mgdl"  "crp_ngml"          "height_m"          "
```

```r
# drop the helper/intermediate columns we created
clinical_data <- clinical_data |>
  select(-visit_year_month)

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"        "visit_year"        "visit_month
 [9] "smoking"           "treatment"         "stage"             "weight_kg"         "height_cm"
[17] "cholesterol_mgdl"  "crp_ngml"          "height_m"          "BMI"               "BMI_class"
```

THIS SHOULD BE ONLY ON THE SITE, NOT IN STUDENTS MARKDOWN `select` also supports handy helper functions:

```r
# select all columns that start with "diag"
clinical_data |> select(starts_with("visit")) |> head()
```

```
       visit_date visit_year visit_month visit_day
PT-001 2020-03-23       2020          03        23
PT-002 2023-10-31       2023          10        31
PT-003 2023-06-27       2023          06        27
PT-004 2018-10-25       2018          10        25
PT-005 2018-02-19       2018          02        19
PT-006 2023-02-06       2023          02        06
```

```r
# select all columns that contain "bp"
clinical_data |> select(contains("bp")) |> head()
```

```
       systolic_bp_mmhg diastolic_bp_mmhg
PT-001              143                90
PT-002              138                78
PT-003              144                84
PT-004              122                93
PT-005              127                89
PT-006              142                92
```

2.3.3 Boolean Operators

A **boolean** (or logical) is a data type that can only take two values: `TRUE` or `FALSE`. It is the backbone of filtering and conditional logic in R.

A boolean is the answer to a yes/no question.

For example: *"Is this patient in stage III or IV?"* → `TRUE` or `FALSE`.

2.3.3.1 Comparison operators

These operators compare two values and return a boolean:

Operator	Meaning	Example	Result
`==`	equal to	`3 == 3`	`TRUE`
`!=`	not equal to	`3 != 4`	`TRUE`
`>`	greater than	`5 > 3`	`TRUE`
`<`	less than	`2 < 1`	`FALSE`
`>=`	greater or equal	`3 >= 3`	`TRUE`
`<=`	less or equal	`2 <= 1`	`FALSE`

```
# simple examples
print(10 > 5)
```

```
[1] TRUE
```

```
print(10 == 10)
```

```
[1] TRUE
```

```
print(10 != 10)
```

```
[1] FALSE
```

```
print(3.14 >= 3.14)
```

```
[1] TRUE
```

When applied to a **vector**, the comparison is done **element-wise** and returns a boolean vector:

```
ages <- clinical_data$age
head(ages)
```

```
[1] 66 62 54 64 51 55
```

```
is_over_60 <- ages > 60
head(is_over_60)
```

```
[1]  TRUE  TRUE FALSE  TRUE FALSE FALSE
```

Under the hood, TRUE is stored as a 1, while FALSE as 0. This is useful when we want to get the number of TRUE values of a vector. How? By summing up all the values:

```r
sum(is_over_60)
```

```
[1] 97
```

2.3.3.2 Logical operators

We often need to combine multiple conditions. This is done with **logical operators**:

Operator	Meaning	Example
`&`	AND — both conditions must be true	`age > 50 & sex == "Female"`
`|`	OR — at least one condition must be true	`stage == "III" | stage == "IV"`
`!`	NOT — negates a condition	`!is.na(crp_ngml)`

```r
# patients who are Female AND over 60
female_over60 <- (clinical_data$sex == "Female") & (clinical_data$age > 60)
sum(female_over60)
```

```
[1] 46
```

```r
# patients in advanced stage (III or IV)
advanced_stage <- (clinical_data$stage == "III") | (clinical_data$stage == "IV")
sum(advanced_stage)
```

```
[1] 86
```

```r
# NOT: patients who are NOT in the Control group
not_control <- clinical_data$treatment != "Control"
sum(not_control)
```

```
[1] 159
```

```
# equivalent using !
not_control_2 <- !(clinical_data$treatment == "Control")
sum(not_control_2)
```

```
[1] 159
```

A very useful operator when checking membership in a list of values is `%in%`:

```
# patients from Milan or Rome (equivalent to == "Milan" | == "Rome")
big_sites <- clinical_data$site %in% c("Milan", "Rome")
sum(big_sites)
```

```
[1] 170
```

::: .{exercise #bool-sum}

How many patients have been diagnosed as stage I, in Turin in 2018?

:::

Solution

```
stageI_turin_2018 <- (clinical_data$site == "Turin") & (clinical_data$stage == "I") &
sum(stageI_turin_2018)
```

```
[1] 3
```

2.3.4 filter and filter out

`filter` keeps only the **rows** that satisfy one or more conditions:

```
# Keep only patients treated with Drug_A
drug_a_patients <- clinical_data |>
  filter(treatment == "Drug_A")

nrow(drug_a_patients)
```

```
[1] 85
```

```r
head(drug_a_patients)
```

```
       patient_id  site visit_date visit_year visit_month visit_day age    sex smoking treatment
PT-004     PT-004 Milan 2018-10-25       2018          10        25  64   Male   never    Drug_A
PT-009     PT-009 Milan 2020-03-10       2020          03        10  61   Male current    Drug_A
PT-012     PT-012  Rome 2020-10-13       2020          10        13  74   Male   never    Drug_A
PT-019     PT-019  Rome 2020-08-26       2020          08        26  66 Female current    Drug_A
PT-026     PT-026 Turin 2023-04-10       2023          04        10  49 Female current    Drug_A
PT-031     PT-031 Turin 2025-10-05       2025          10        05  76   Male current    Drug_A
       cholesterol_mgdl crp_ngml height_m  BMI   BMI_class
PT-004              129     1.63    1.782 24.9 Underweight
PT-009              173     3.31    1.699 30.6      Normal
PT-012              208     5.75    1.651 33.6      Normal
PT-019              203     0.75    1.600 27.3  Overweight
PT-026              195     5.23    1.633 21.6 Underweight
PT-031              245    19.73    1.791 28.2  Overweight
```

```r
# Patients in advanced stage AND not in control group
advanced_treated <- clinical_data |>
  filter((stage == "III" | stage == "IV") & treatment != "Control")

nrow(advanced_treated)
```

```
[1] 58
```

```r
# Using %in% for multiple values
milan_or_rome <- clinical_data |>
  filter(site %in% c("Milan", "Rome"))

nrow(milan_or_rome)
```

```
[1] 170
```

To **filter out** (i.e., *exclude* rows matching a condition), use `filter_out`:

```r
# Exclude patients from Naples
no_naples <- clinical_data |>
  filter_out(site %in% c("Naples"))

nrow(no_naples)
```

```
[1] 240
```

Pipes shine when you chain multiple operations:

```r
# Female patients, over 55, from Milan, keep only key columns
subset_females <- clinical_data |>
  filter(sex == "Female", age > 55, site == "Milan") |>
  select(patient_id, age, stage, treatment, BMI, glucose_mgdl)

head(subset_females)
```

```
       patient_id age stage treatment  BMI glucose_mgdl
PT-008     PT-008  63   III   Control 24.3           82
PT-014     PT-014  59    II   Control 29.8           91
PT-029     PT-029  61   III   Control 24.3           66
PT-037     PT-037  59    II   Control 15.2           92
PT-051     PT-051  75     I    Drug_B 26.2           77
PT-053     PT-053  62    II    Drug_A 26.7           95
```

```r
nrow(subset_females)
```

```
[1] 28
```

Note: inside `filter`, multiple conditions separated by commas are treated as AND.

Exercise 2.2. From the full dataset, extract patients who:

- are in stage II or III,
- have a glucose level between 90 and 130 mg/dL (inclusive),
- are assigned to Drug_A or Drug_B.

Keep only `patient_id`, `site`, `stage`, `treatment`, and `glucose_mgdl`.

Solution

```r
exercise_result <- clinical_data |>
  filter(stage %in% c("II", "III"),
         glucose_mgdl >= 90,
         glucose_mgdl <= 130,
         treatment %in% c("Drug_A", "Drug_B")) |>
  select(patient_id, site, stage, treatment, glucose_mgdl)

head(exercise_result)
```

```
       patient_id  site stage treatment glucose_mgdl
PT-011     PT-011  Rome    II    Drug_B          119
PT-012     PT-012  Rome    II    Drug_A          101
PT-020     PT-020 Milan    II    Drug_B          115
PT-021     PT-021  Rome   III    Drug_B          122
PT-022     PT-022 Turin   III    Drug_B          127
PT-026     PT-026 Turin    II    Drug_A          106
```

```r
nrow(exercise_result)
```

```
[1] 55
```

2.3.5 group_by, summarize and reframe

One of the most powerful operations in data analysis is **grouping** data and computing summaries per group. In R, this is done with `group_by` + `summarize`.

How it works:

1. `group_by(column)` tells R: *"split the dataframe into groups based on the values of this column"*
2. `summarize(...)` computes one summary value per group

```
Original data:                 After group_by(treatment):

patient  treatment  glucose       GROUP: Control
PT-001   Control    105           PT-001  Control  105
PT-002   Drug_A     118    --->   PT-007  Control  98
PT-003   Drug_B     97            ...
PT-007   Control    98     --->   GROUP: Drug_A
...                               PT-002  Drug_A  118
                                  ...
```

We can perform several operation with this combo, and answer many questions. For example "*Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?*".

```r
# Mean glucose per treatment group
glucose_by_treatment <- clinical_data |>
  group_by(treatment) |>
  summarize(mean_glucose = mean(glucose_mgdl),
```

```
            sd_glucose   = sd(glucose_mgdl),
            n_patients   = n())

glucose_by_treatment
```

```
# A tibble: 3 x 4
  treatment mean_glucose sd_glucose n_patients
  <fct>            <dbl>      <dbl>      <int>
1 Control           104.       21.5         81
2 Drug_A            107.       23.6         85
3 Drug_B            101.       24.1         74
```

You can also group by multiple columns at once.

"*For each treatment/sex combination, which is the mean BMI and the median of the age?*"

```
# Summary by treatment AND sex
summary_by_group <- clinical_data |>
  group_by(treatment, sex) |>
  summarize(mean_BMI    = round(mean(BMI), 2),
            median_age  = median(age),
            n           = n(),
            .groups = "drop")   # good practice: drop grouping after summarize

summary_by_group
```

```
# A tibble: 6 x 5
  treatment sex    mean_BMI median_age     n
  <fct>     <fct>     <dbl>      <dbl> <int>
1 Control   Female     25.9       55.5    40
2 Control   Male       25.2       55      41
3 Drug_A    Female     25.5       58      41
4 Drug_A    Male       26.1       60      44
5 Drug_B    Female     25.2       61      37
6 Drug_B    Male       26.6       57      37
```

METTERE SOLO NEL SITO

`summarize` returns **one row per group**. If you need to return **multiple rows per group** (e.g., all quantiles), use `reframe` instead:

```
# Get quartiles of CRP per treatment group
crp_quantiles <- clinical_data |>
  group_by(treatment) |>
```

```
  reframe(quantile_value = quantile(crp_ngml, probs = c(0.25, 0.5, 0.75)),
          quantile_name  = c("Q1", "median", "Q3"))

crp_quantiles
```

```
# A tibble: 9 x 3
  treatment quantile_value quantile_name
  <fct>              <dbl> <chr>
1 Control             2.24 Q1
2 Control             3.44 median
3 Control             5.77 Q3
4 Drug_A              2.19 Q1
5 Drug_A              3.03 median
6 Drug_A              5.29 Q3
7 Drug_B              1.99 Q1
8 Drug_B              4.01 median
9 Drug_B              5.79 Q3
```

Another useful function: `count`, which is a shortcut for `group_by` + `summarize(n = n())`:

```
# Count patients per site and stage
clinical_data |>
  count(site, stage, sort = TRUE)
```

```
    site stage  n
1  Milan    II 40
2  Milan     I 34
3   Rome    II 24
4  Milan   III 23
5  Turin     I 22
6  Turin   III 22
7  Turin    II 19
8   Rome     I 15
9   Rome   III 13
10 Milan    IV 12
11  Rome    IV  9
12 Turin    IV  7
```

FINO A QUI

Exercise 2.3. For each combination of `site` and `sex`, compute: - the number of patients, - the mean systolic blood pressure, - the proportion of hypertensive patients (hint: hypertension is when systolic blood pression is > 140).

Solution

```
bp_summary <- clinical_data |>
    mutate(hypertension = systolic_bp_mmhg > 140) |>
    group_by(site, sex) |>
    summarize(
        n_patients          = n(),
        mean_systolic_bp_mmhg    = round(mean(systolic_bp_mmhg), 1),
        prop_hypertensive   = round(mean(hypertension), 3),
        .groups = "drop"
    )

bp_summary
```

```
# A tibble: 6 x 5
  site  sex    n_patients mean_systolic_bp_mmhg prop_hypertensive
  <fct> <fct>       <int>                 <dbl>             <dbl>
1 Milan Female         51                  130              0.333
2 Milan Male           58                  134.             0.362
3 Rome  Female         26                  131.             0.346
4 Rome  Male           35                  132.             0.257
5 Turin Female         41                  132.             0.317
6 Turin Male           29                  127.             0.207
```

2.3.6 Join / Merge

In real-world analyses, data about the same subjects often lives in **separate files**: demographic data here, lab results there, outcomes somewhere else. To combine them, we **join** them on a shared key column (usually a patient ID).

Let's load the separate lab results file:

```
lab_data <- read.csv("data/lab_results.csv")
str(lab_data)
```

```
'data.frame':   160 obs. of  6 variables:
 $ patient_id     : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ hemoglobin_gdl : num  14.7 12.3 11.8 14.1 12.1 10.4 11.7 14.5 13.9 8.1 ...
 $ wbc_10e9l      : num  5.02 9.64 8.9 10.8 7.69 ...
 $ platelets_10e9l: int  318 138 83 256 283 153 237 233 166 220 ...
 $ alt_ul         : int  28 27 39 15 34 32 37 30 30 25 ...
 $ creatinine_mgdl: num  0.67 0.68 1.1 1.02 0.38 1.26 1.1 0.76 1.26 1 ...
```

```r
head(lab_data)
```

```
  patient_id hemoglobin_gdl wbc_10e9l platelets_10e9l alt_ul creatinine_mgdl
1     PT-001           14.7      5.02             318     28            0.67
2     PT-002           12.3      9.64             138     27            0.68
3     PT-003           11.8      8.90              83     39            1.10
4     PT-004           14.1     10.80             256     15            1.02
5     PT-005           12.1      7.69             283     34            0.38
6     PT-006           10.4     11.34             153     32            1.26
```

The key column is `patient_id`. Not all patients have lab results. Depending on how we want to handle this, we choose different types of join:

Join type	What it does
`left_join(x, y)`	Keep **all rows from x**, add y columns where available (NA if no match)
`right_join(x, y)`	Keep all rows from y
`inner_join(x, y)`	Keep only rows that exist in **both** x and y
`full_join(x, y)`	Keep all rows from both, NA where no match

Let's see all approaches:

```r
# left_join: keep all patients, add lab data where available
full_with_labs <- clinical_data |>
  left_join(lab_data, by = "patient_id")

nrow(full_with_labs)
```

```
[1] 240
```

```r
# patients without lab data will have NA in lab columns
sum(is.na(full_with_labs$hemoglobin_gdl))
```

```
[1] 80
```

```r
summary(full_with_labs)
```

```
 patient_id           site      visit_date         visit_year        visit_month
Length:240         Milan:109   Length:240         Length:240         Length:240
Class :character   Rome : 61   Class :character   Class :character   Class :character
Mode  :character   Turin: 70   Mode  :character   Mode  :character   Mode  :character
```

```
  weight_kg        height_cm     systolic_bp_mmhg diastolic_bp_mmhg  glucose_mgdl
Min.   : 38.40   Min.   :147.4   Min.   : 89.0    Min.   : 52.00    Min.   : 43.00
1st Qu.: 66.33   1st Qu.:162.8   1st Qu.:117.0    1st Qu.: 75.00    1st Qu.: 89.75
Median : 73.25   Median :168.5   Median :130.5    Median : 82.00    Median :104.00
Mean   : 73.60   Mean   :169.3   Mean   :131.4    Mean   : 82.02    Mean   :104.43
3rd Qu.: 82.60   3rd Qu.:176.2   3rd Qu.:143.2    3rd Qu.: 89.25    3rd Qu.:119.00
Max.   :104.40   Max.   :193.4   Max.   :195.0    Max.   :108.00    Max.   :181.00

hemoglobin_gdl    wbc_10e9l      platelets_10e9l     alt_ul       creatinine_mgdl
Min.   : 8.10   Min.   : 1.900   Min.   : 43.0   Min.   :  9.00   Min.   :0.3800
1st Qu.:12.57   1st Qu.: 5.985   1st Qu.:179.5   1st Qu.: 20.00   1st Qu.:0.7600
Median :13.90   Median : 6.925   Median :224.5   Median : 29.50   Median :0.9350
Mean   :13.54   Mean   : 7.106   Mean   :220.2   Mean   : 34.97   Mean   :0.9321
3rd Qu.:14.70   3rd Qu.: 8.430   3rd Qu.:259.2   3rd Qu.: 44.00   3rd Qu.:1.0800
Max.   :19.00   Max.   :12.230   Max.   :349.0   Max.   :156.00   Max.   :1.6500
NA's   :80      NA's   :80       NA's   :80      NA's   :80       NA's   :80
```

```r
# inner_join: keep only the 90 patients who have both clinical and lab data
matched_only <- clinical_data |>
  inner_join(lab_data, by = "patient_id")

nrow(matched_only)
```

```
[1] 160
```

In most clinical research scenarios, `left_join` is the most common choice: you start with your full patient list and enrich it with additional data, accepting that some patients may lack certain measurements (NAs).

Exercise 2.4.

1. Perform a `left_join` of `clinical_data` with `lab_data`.
2. Among patients who have lab data available, compute the mean hemoglobin and mean creatinine per treatment group.
3. Which group has the highest mean creatinine?

Solution

```r
# 1. Left join
joined_df <- clinical_data |>
  left_join(lab_data, by = "patient_id")

# 2. Summarize (filter out NA first, or use na.rm = TRUE)
lab_summary <- joined_df |>
  filter(!is.na(hemoglobin_gdl)) |>
  group_by(treatment) |>
  summarize(mean_hemoglobin   = round(mean(hemoglobin_gdl), 2),
            mean_creatinine   = round(mean(creatinine_mgdl), 3),
            n                 = n(),
            .groups = "drop")

lab_summary
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.4           0.956    59
2 Drug_A               13.6           0.911    55
3 Drug_B               13.6           0.927    46
```

```r
# 3. Highest mean creatinine
lab_summary |> arrange(desc(mean_creatinine))
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.4           0.956    59
2 Drug_B               13.6           0.927    46
3 Drug_A               13.6           0.911    55
```

2.4 Save your results

After all this work, let's save the enriched and cleaned dataset:

```r
# Save the left-joined dataset with all derived variables
final_data <- clinical_data |>
  left_join(lab_data, by = "patient_id")
```

```r
write.csv(final_data,
          file = "output/clinical_data_enriched.csv",
          row.names = FALSE)
```

2.5 Home exercise

Try to do the following on your own:

1. Load `clinical_data_enriched.csv` from the output folder.
2. Clean up: keep only the columns `patient_id`, `site`, `sex`, `age`, `treatment`, `stage`, `BMI`, `BMI_class`, `glucose_mgdl`, `cholesterol_mgdl`, `crp_ngml`, `hemoglobin_gdl`, `creatinine_mgdl`.
3. Restrict to patients who have complete data (no NAs) — hint: use function `drop_na()`, see this.
4. For each stage, compute: number of patients, mean age, mean BMI, and proportion of obese patients.
5. Save the summary table to `output/summary_by_stage.csv`.

```r
library(tidyverse)

# 1. Load the enriched dataset
clinical_data_enriched <- read.csv("output/clinical_data_enriched.csv")

# 2. Keep only the columns of interest
clean_data <- clinical_data_enriched |>
  select(patient_id, site, sex, age, treatment, stage,
         BMI, BMI_class, glucose_mgdl, cholesterol_mgdl,
         crp_ngml, hemoglobin_gdl, creatinine_mgdl)

# 3. Restrict to patients with complete data (no NAs)
clean_data <- clean_data |>
  drop_na()

cat("Patients with complete data:", nrow(clean_data), "\n")
```

```
Patients with complete data: 160
```

```r
# 4. Summary by stage
summary_by_stage <- clean_data |>
  group_by(stage) |>
```

```r
  summarize(
    n_patients       = n(),
    mean_age         = round(mean(age), 1),
    mean_BMI         = round(mean(BMI), 2),
    prop_obese       = round(mean(BMI_class == "Obese"), 3),
    .groups = "drop"
  )

print(summary_by_stage)
```

```
# A tibble: 4 x 5
  stage n_patients mean_age mean_BMI prop_obese
  <chr>      <int>    <dbl>    <dbl>      <dbl>
1 I             42     59.3     26.4      0.048
2 II            61     58.1     25.9      0.066
3 III           39     55.7     25.3      0.026
4 IV            18     57.2     27.7      0
```

```r
# 5. Save
write.csv(summary_by_stage,
          file = "output/summary_by_stage.csv",
          row.names = FALSE)
```

2.6 Bonus: how the clinical data were created

If you want to understand how we generated the dataset for this lesson, download the R script Notice the use of `set.seed` to make results reproducible.

Chapter 3

Data visualization

Data visualization is a crucial step of data analysis, as it can help us in:

- Understanding the distribution of our data
- Evaluate outliers
- Visualize immediately the differences between groups
- Communicate our results to peers

I suggest you to visualize data as soon as you have finished to prepare them, prior to any statistical test (even prior to test for parametric assumptions), that's why today we are doing it.

3.1 Data visualization in R with ggplot2

`ggplot2` is a package built for data visualization in R. It is simple to use and it works mainly on dataframes.

To create a plot, we need to use different functions, that act as single building blocks to change single aspects of the plot. Here an example of mock code to create a plot with all the functions:

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>() +
  facet_<type>() +
  coord_<type>() +
  theme_<type>
```

But, do we need all of them? No, the basic plots can be created with just `ggplot() + geom_<type>()`.

3.2 Geom

The `geom` function specifies the "geometry" we want to give to the data: basically, which type of graph we want (barplot, scatter, histogram, etc). It is the necessary and sufficient function to create a plot.

Here the list of all possible geom to use with the basic package.

First of all, let's load the data from previous session, and keep only the rows with all the information:

```r
suppressPackageStartupMessages(library(tidyverse))
df <- read.csv("output/clinical_data_enriched.csv", header = T, stringsAsFactors = T)
df <- na.omit(df)
str(df)
```

```
'data.frame':   160 obs. of  26 variables:
 $ patient_id       : Factor w/ 240 levels "PT-001","PT-002",..: 1 2 3 4 5 6 7 9 10 11
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 1 1 3 1 1 1 1 1 1 2 ...
 $ visit_date       : Factor w/ 240 levels "2018-01-11","2018-01-18",..: 72 180 150 51
 $ visit_year       : int  2020 2023 2023 2018 2018 2023 2025 2020 2018 2025 ...
 $ visit_month      : int  3 10 6 10 2 2 2 3 4 10 ...
 $ visit_day        : int  23 31 27 25 19 6 3 10 17 23 ...
 $ age              : int  66 62 54 64 51 55 71 61 55 69 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 2 2 1 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 2 1 1 3 2 3 2 1 1 1 ..
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 1 3 3 2 3 1 1 2 1 3 ..
 $ stage            : Factor w/ 4 levels "I","II","III",..: 1 2 3 4 1 2 1 3 2 2 ...
 $ weight_kg        : num  78.8 87.6 61.6 79.2 96.1 69 74.1 88.2 71 59.2 ...
 $ height_cm        : num  181 170 162 178 182 ...
 $ systolic_bp_mmhg : int  143 138 144 122 127 142 125 105 179 99 ...
 $ diastolic_bp_mmhg: int  90 78 84 93 89 92 62 80 75 76 ...
 $ glucose_mgdl     : int  102 53 80 103 123 110 119 138 131 119 ...
 $ cholesterol_mgdl : int  250 207 180 129 178 173 239 173 204 200 ...
 $ crp_ngml         : num  1.37 11.09 0.81 1.63 1.25 ...
 $ height_m         : num  1.81 1.7 1.62 1.78 1.82 ...
 $ BMI              : num  23.9 30.3 23.4 24.9 28.9 25.1 26.6 30.6 22.2 21.6 ...
 $ BMI_class        : Factor w/ 4 levels "Normal","Obese",..: 4 1 4 4 3 3 3 1 4 4 ...
 $ hemoglobin_gdl   : num  14.7 12.3 11.8 14.1 12.1 10.4 11.7 14.5 13.9 8.1 ...
 $ wbc_10e9l        : num  5.02 9.64 8.9 10.8 7.69 ...
 $ platelets_10e9l  : int  318 138 83 256 283 153 237 233 166 220 ...
 $ alt_ul           : int  28 27 39 15 34 32 37 30 30 25 ...
```

```
 $ creatinine_mgdl  : num  0.67 0.68 1.1 1.02 0.38 1.26 1.1 0.76 1.26 1 ...
 - attr(*, "na.action")= 'omit' Named int [1:80] 8 16 17 24 27 28 29 32 33 34 ...
  ..- attr(*, "names")= chr [1:80] "8" "16" "17" "24" ...
```

```r
summary(df)
```

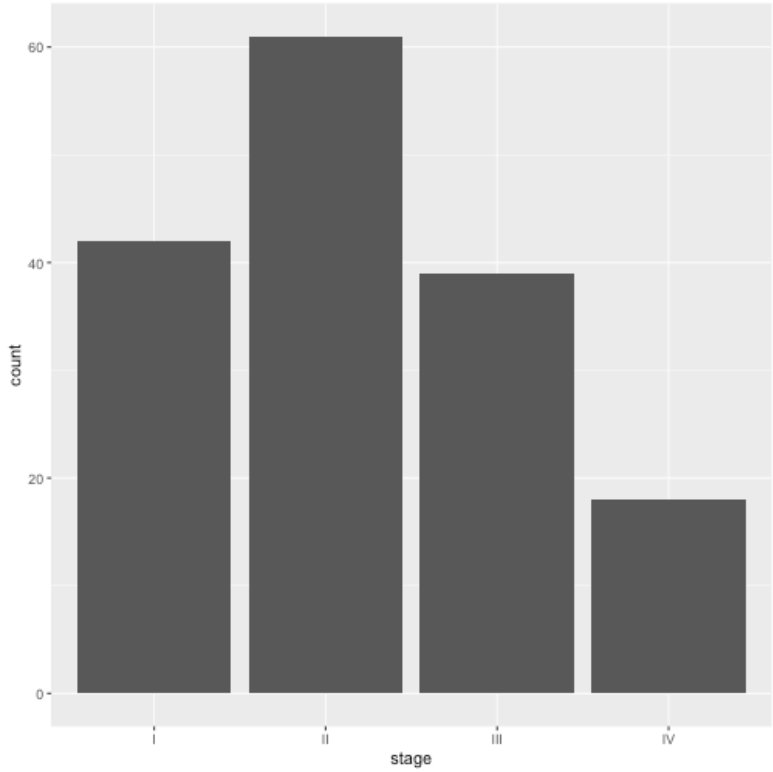
```
   patient_id     site         visit_date    visit_year    visit_month       visit_day          a
 PT-001 :  1   Milan:75   2018-01-11:  1   Min.   :2018   Min.   : 1.000   Min.   : 1.00   Min.
 PT-002 :  1   Rome :42   2018-01-18:  1   1st Qu.:2020   1st Qu.: 3.000   1st Qu.: 8.75   1st Qu
 PT-003 :  1   Turin:43   2018-01-19:  1   Median :2023   Median : 6.000   Median :16.00   Median
 PT-004 :  1              2018-01-20:  1   Mean   :2022   Mean   : 5.969   Mean   :16.14   Mean
 PT-005 :  1              2018-01-31:  1   3rd Qu.:2025   3rd Qu.: 9.000   3rd Qu.:24.00   3rd Qu
 PT-006 :  1              2018-02-07:  1   Max.   :2025   Max.   :11.000   Max.   :31.00   Max.
 (Other):154              (Other)   :154
   height_cm     systolic_bp_mmhg diastolic_bp_mmhg  glucose_mgdl   cholesterol_mgdl    crp_ngml
 Min.   :147.4   Min.   : 89.0    Min.   : 52.00    Min.   : 50.0   Min.   :103.0    Min.   : 0.2
 1st Qu.:162.7   1st Qu.:115.8    1st Qu.: 75.00    1st Qu.: 91.0   1st Qu.:178.8    1st Qu.: 2.0
 Median :168.6   Median :128.0    Median : 81.50    Median :106.0   Median :200.5    Median : 3.5
 Mean   :169.0   Mean   :129.9    Mean   : 81.96    Mean   :106.2   Mean   :198.8    Mean   : 4.5
 3rd Qu.:175.3   3rd Qu.:143.0    3rd Qu.: 90.00    3rd Qu.:121.2   3rd Qu.:221.0    3rd Qu.: 5.7
 Max.   :193.4   Max.   :179.0    Max.   :108.00    Max.   :181.0   Max.   :295.0    Max.   :28.1

   wbc_10e9l      platelets_10e9l     alt_ul       creatinine_mgdl
 Min.   : 1.900   Min.   : 43.0   Min.   :  9.00   Min.   :0.3800
 1st Qu.: 5.985   1st Qu.:179.5   1st Qu.: 20.00   1st Qu.:0.7600
 Median : 6.925   Median :224.5   Median : 29.50   Median :0.9350
 Mean   : 7.106   Mean   :220.2   Mean   : 34.97   Mean   :0.9321
 3rd Qu.: 8.430   3rd Qu.:259.2   3rd Qu.: 44.00   3rd Qu.:1.0800
 Max.   :12.230   Max.   :349.0   Max.   :156.00   Max.   :1.6500
```

3.2.1 Barplot

We create a basic barplot to show the number of patient for each stage using the `geom_bar` function:

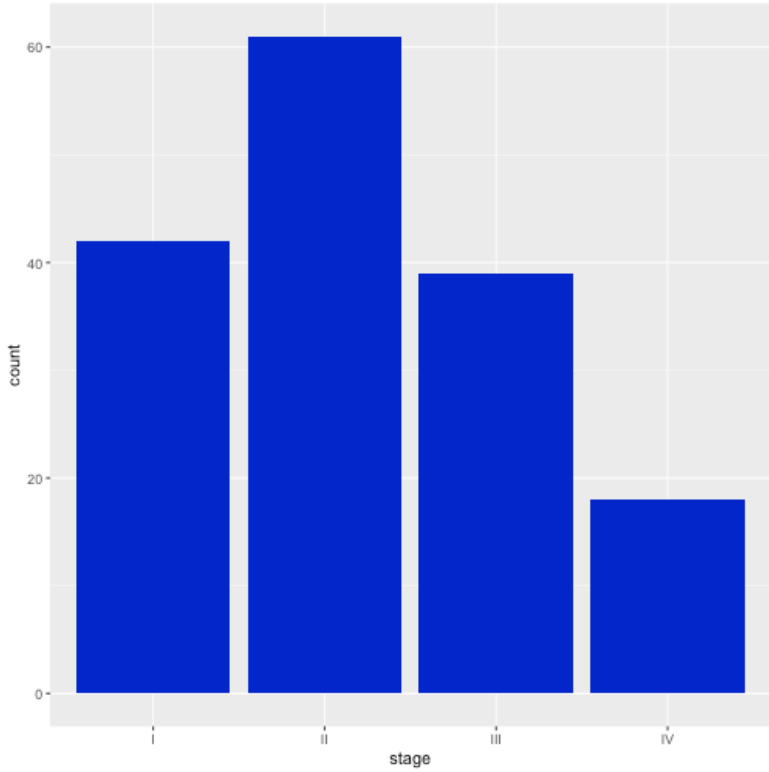
```r
suppressPackageStartupMessages(library(ggplot2))
```

```r
ggplot(df) +
  geom_bar(mapping = aes(x = stage))
```



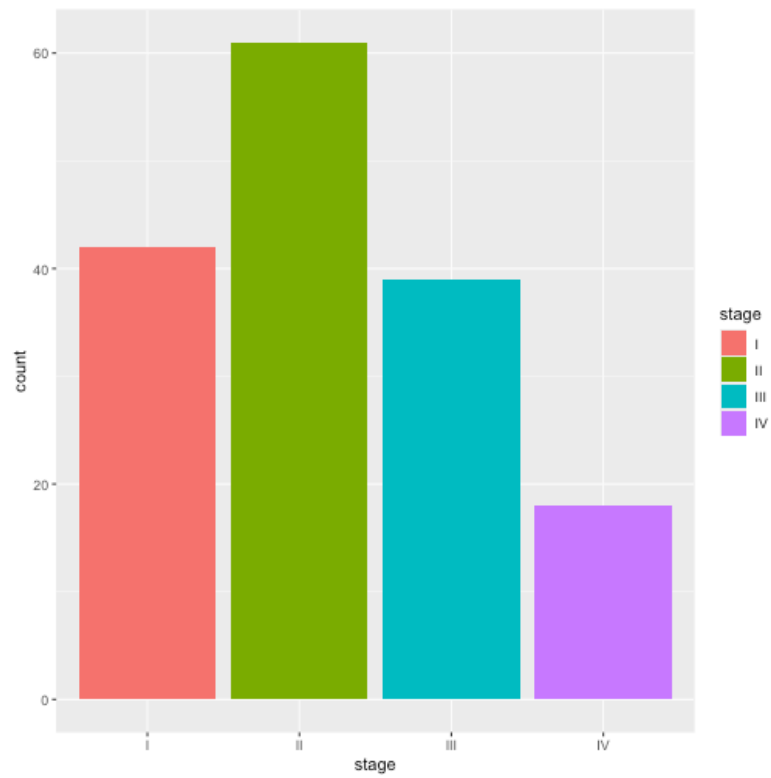
Let's add some colors. We can set colors for all bars:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage), fill = "blue3")
```



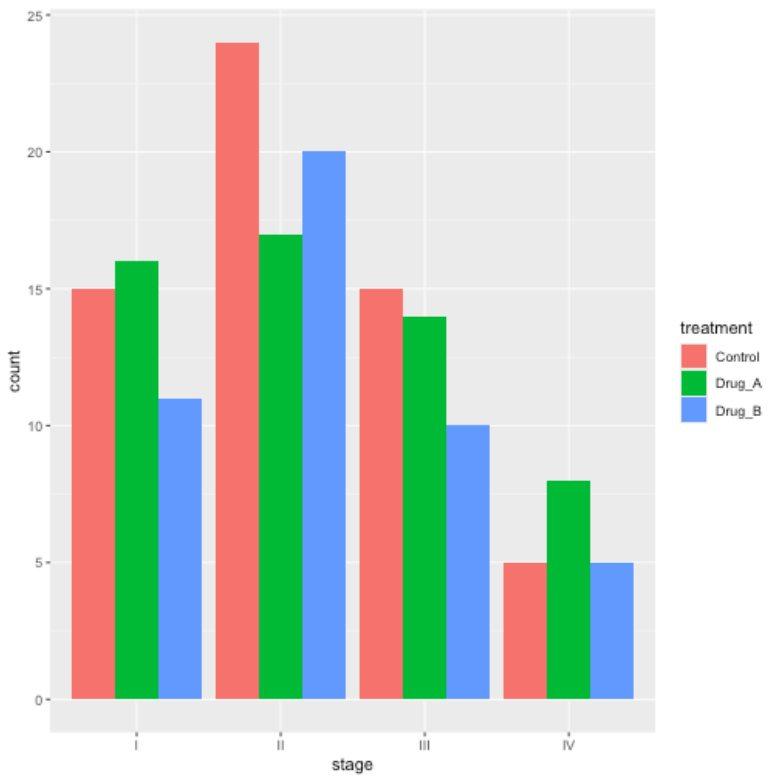
Or based on stage:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = stage))
```



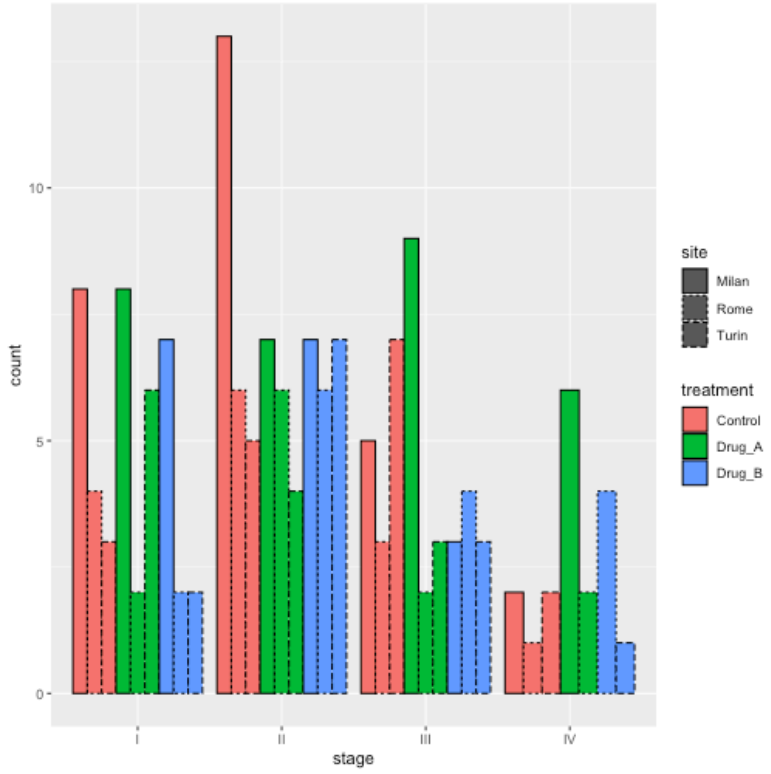
One of the main features of ggplot is that we can easily split data on x axis based on another column. Let's say we want a bar for each treatment within each stage; we can achieve it setting the column in another aes option:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment), position = "dodge")
```



We can set different options in geom bar (both in aes or not), like color (the color of the outline of the bar), linewidth (the size of the outline), linetype (type of the outline of the bars, like dotted or continuous). Here you can find all the options. Here an example:

```
all_n_barplot <- ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment, linetype = site), position = "dodge", color
all_n_barplot
```



3.2.1.1 Stat identity

In this case, geom_bar counts the occurrence of each label of stage and plot them. But what if we have a dataframe with the actual data we want to plot? Let's say that we want to plot the mean values of "diastolic_bp_mmhg" of Stage I females for each treatment; first, we have to create the dataframe for it by filtering the data and calculating the mean:

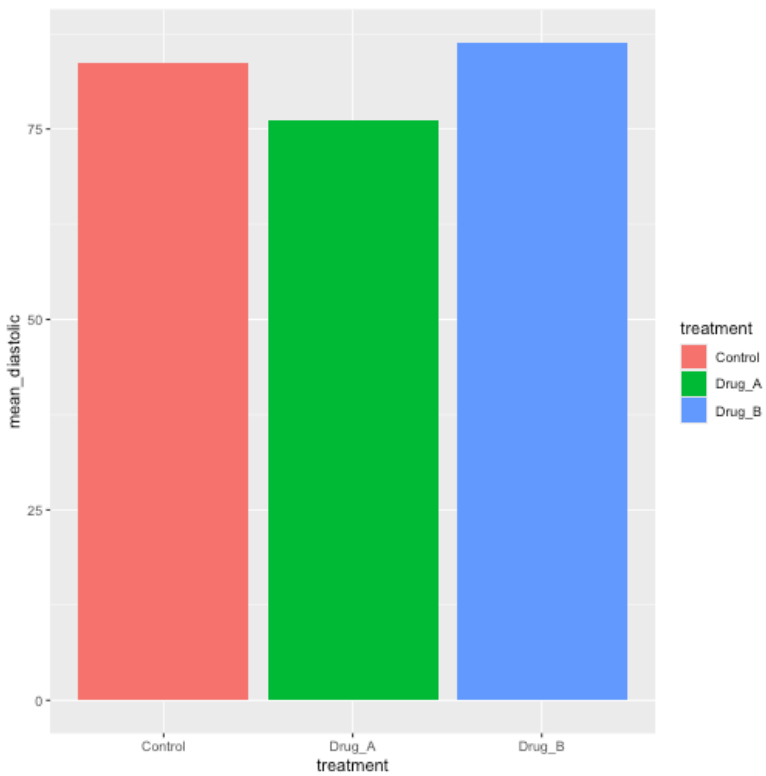
```
stageI_femalse_diastolic_means <- df %>%
  filter(stage == "I" & sex == "Female") %>%
  group_by(treatment) %>%
  summarise("mean_diastolic" = mean(diastolic_bp_mmhg))
stageI_femalse_diastolic_means
```

```
# A tibble: 3 x 2
  treatment mean_diastolic
  <fct>              <dbl>
1 Control             83.6
```

```
2 Drug_A              76.2
3 Drug_B              86.4
```

Then we can use the function specifying both x and y as aes:

```
stageI_female_diastolic_means_barplot <- ggplot(stageI_femalse_diastolic_means) +
  geom_bar(mapping = aes(x = treatment, y = mean_diastolic, fill = treatment), stat = "identity")
stageI_female_diastolic_means_barplot
```

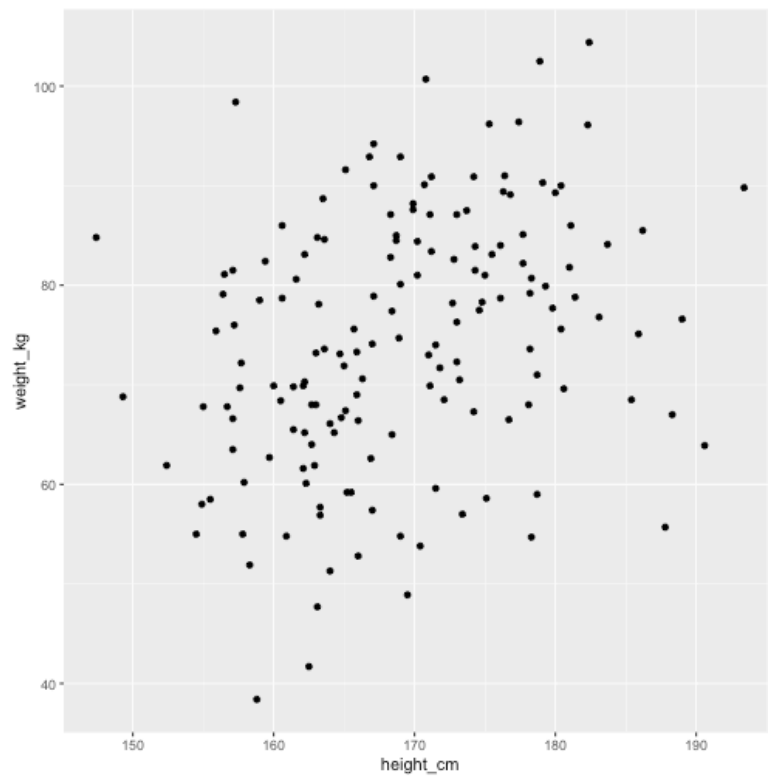


3.2.2 Scatter plot

If we want to create a scatter plot, we can use `geom_point` function.

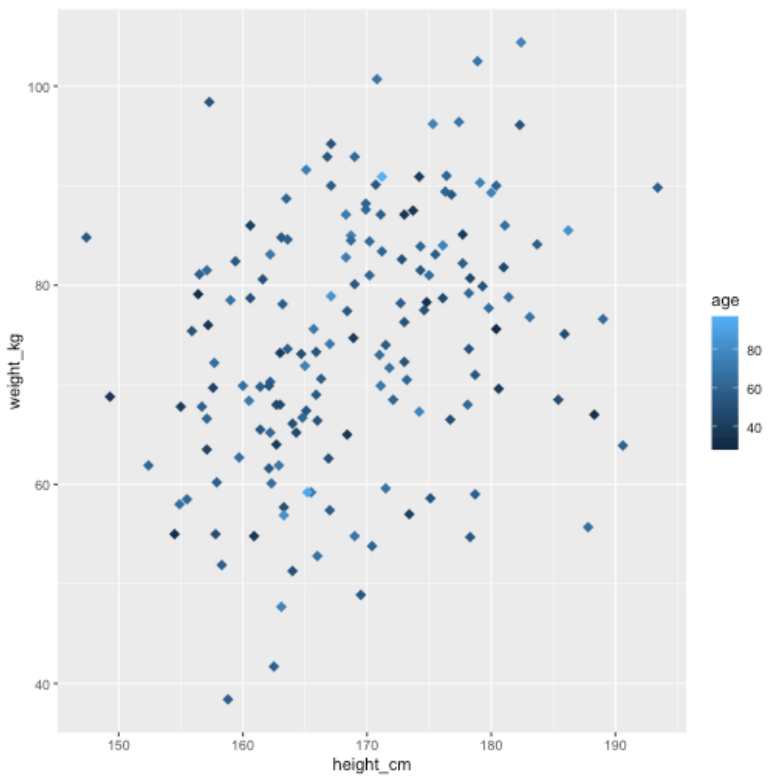
We can create the plot between weight and height:

```
ggplot(df) +
  geom_point(aes(x = height_cm, y = weight_kg))
```



Just from this plot we can see that there is a positive correlation. We can also change color, shape size of the dots. As before, all that is inserted in aes will apply to each dot based on the value of the corresponding column, while outside aes is applied to all dots:

```
corr_plot <- ggplot(df) +
  geom_point(aes(x = height_cm, y = weight_kg, color = age), size = 3, shape = 18)
corr_plot
```

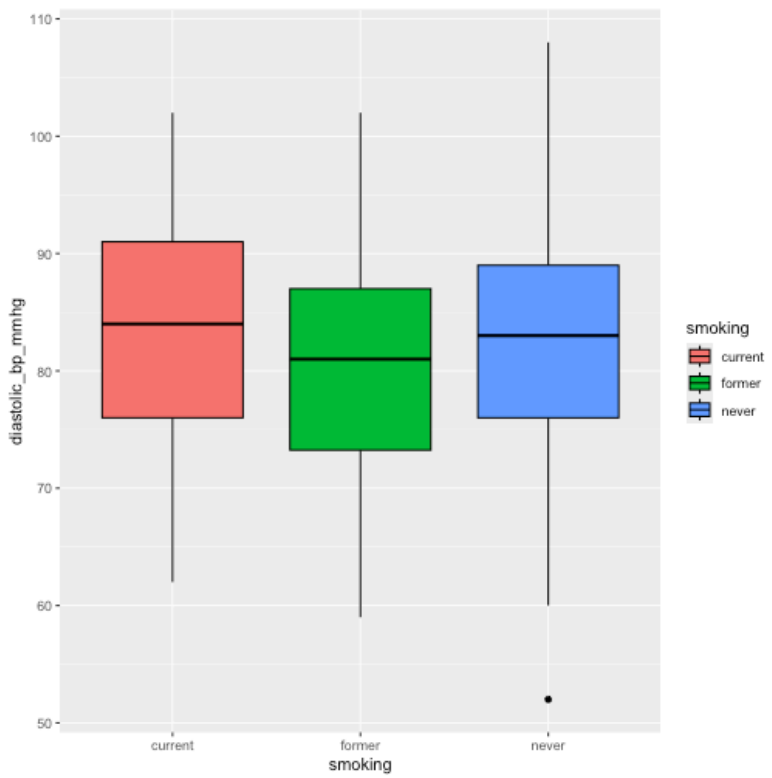


Here the complete list of options for geom_point.

3.2.3 Boxplot

Next, we will see how to create a boxplot, that is one of the most used plot when showing data distribution. Here we look at the diastolic bp in different smoking levels.

```
diastolic_bp_boxplot <- ggplot(df) +
  geom_boxplot(aes(x = smoking, y = diastolic_bp_mmhg, fill = smoking), color = "black", outlier.
diastolic_bp_boxplot
```

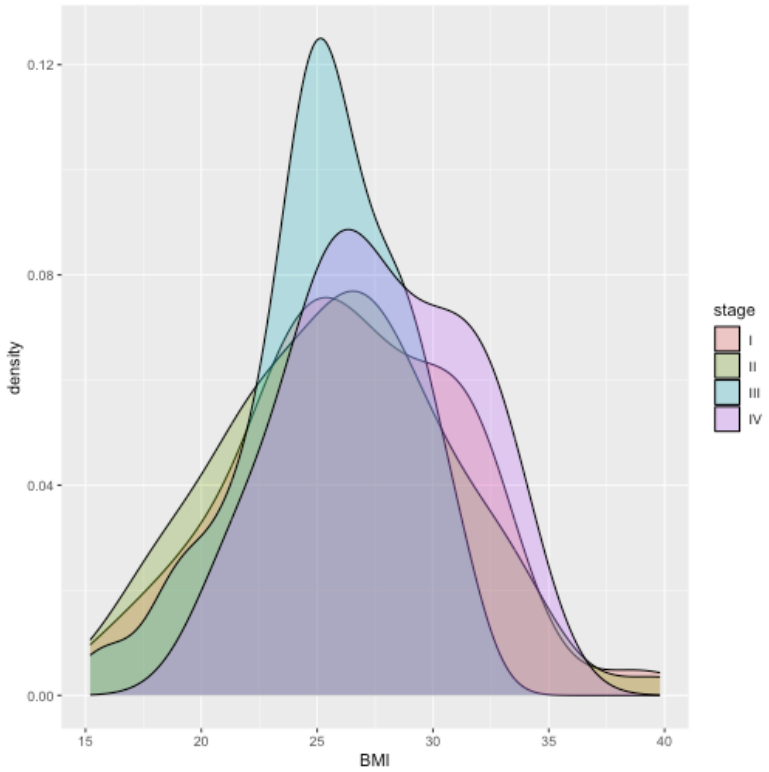


Here the complete list of options for geom_boxplot.

3.2.4 Density plot

Another way to visualize the distribution of the data is the density plot. Let's see the distribution of BMI among stage groups.

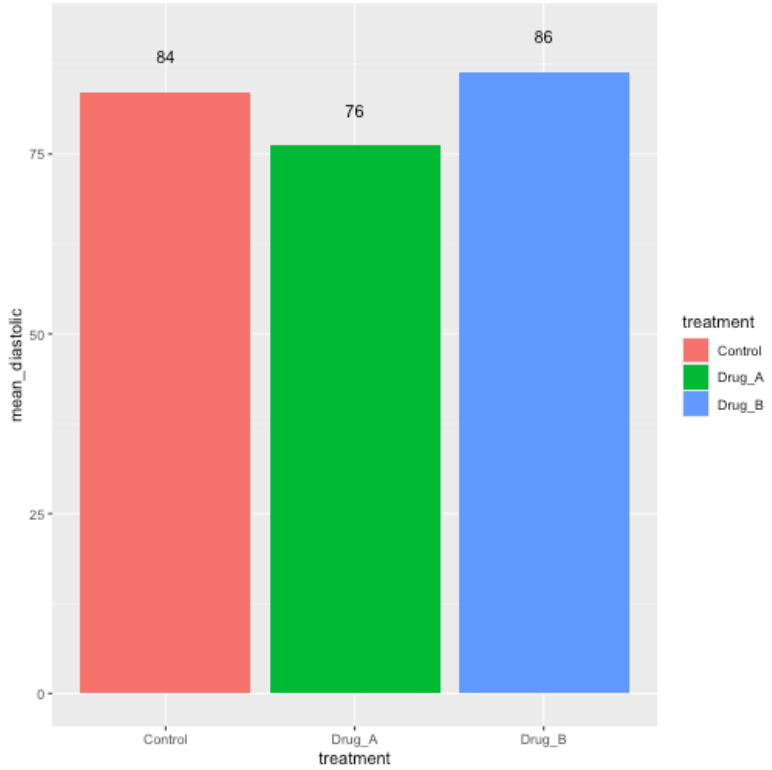
```
BMI_stages_density <- ggplot(df) +
  geom_density(aes(x = BMI, fill = stage), alpha = 0.3)
BMI_stages_density
```



3.2.5 Combine geom

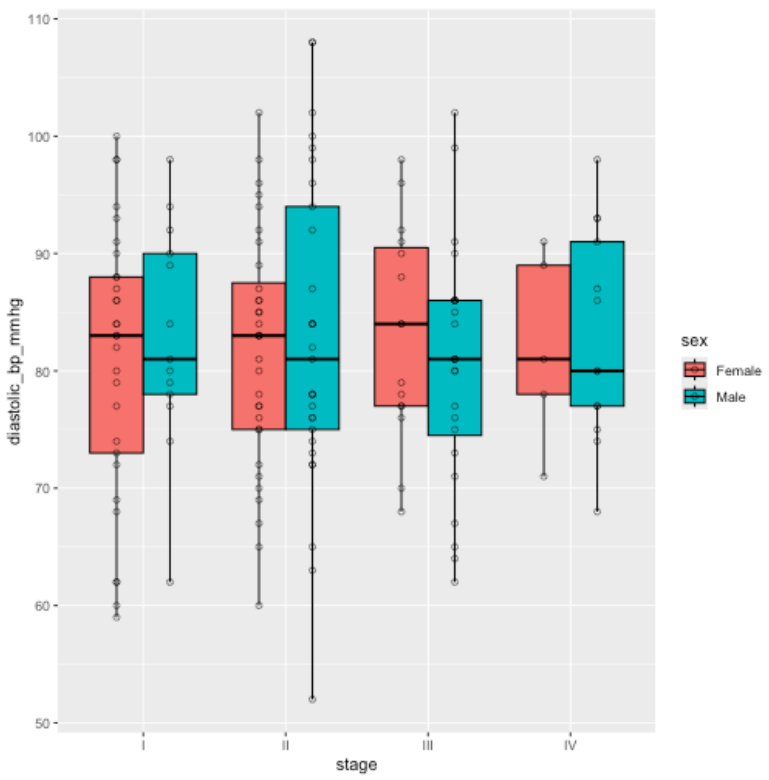
We can also combine geometries. For example, let's add to the *stageI_female_diastolic_means_barplot* the actual values with `geom_text`:

```
stageI_female_diastolic_means_barplot <- stageI_female_diastolic_means_barplot +
  geom_text(aes(x = treatment, y = mean_diastolic, label = round(mean_diastolic)), nudge_y = 5)
stageI_female_diastolic_means_barplot
```



We can combine as many as we want, another example is combine geom_boxplot with geom_point to add individual data points on the plot.

```
ggplot(df) +
  geom_boxplot(aes(x = stage, y = diastolic_bp_mmhg, fill = sex),
               color = "black", outlier.color = "black", position = position_dodge(wid
  geom_jitter(aes(x = stage, y = diastolic_bp_mmhg, fill = sex),
             position = position_dodge(width = 0.75), color = "black", alpha = 0.6, sha
```



3.3 Labs

All these plots are a good starting point, but we can now start to personalize them. The first thing we can change are labs.

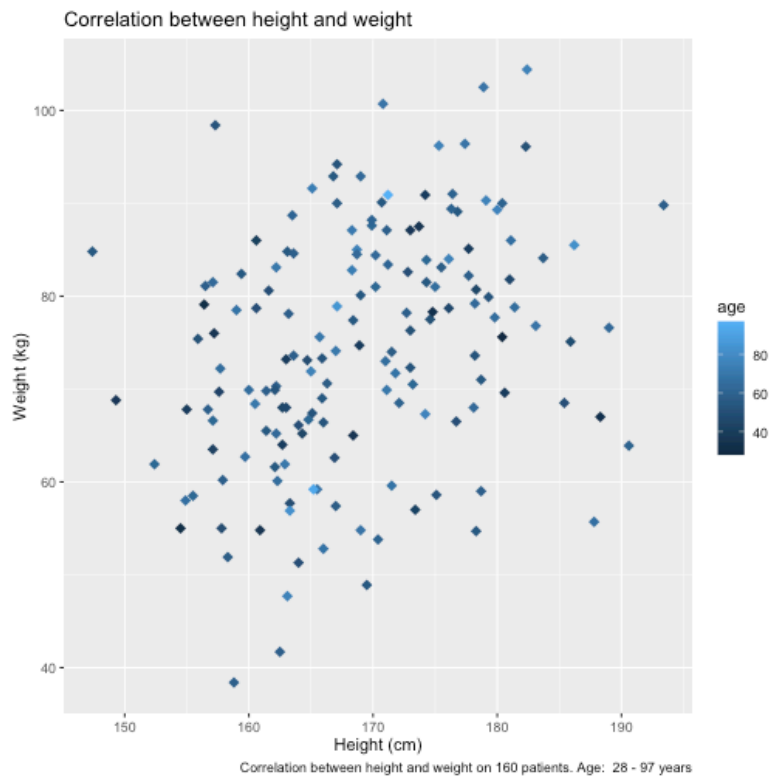
Labs comprehends axis title, plot title and subtitle, legend titles, caption and tag (here the reference page with examples).

```
ggplot() +
  geom_<type>() +
  labs()
```

Let's change the labels of the correlation plot:

```
corr_plot <- corr_plot +
  labs(title = "Correlation between height and weight",
       x = "Height (cm)",
       y = "Weight (kg)",
```

```
        caption = paste("Correlation between height and weight on", dim(df)[1], "patient
                        min(df$age), "-", max(df$age), "years"))
corr_plot
```



3.4 Scales

The definition of scales from the official documentation is "Scales control the details of how data values are translated to visual properties". Practically speaking, with scales we can control the data on the axis, the colors and the other parameters of the aesthetics (linetype, size, fill) by calling the correspondent function (that's what "" meant in the mock code).

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>()
```

We will see the main application of it.

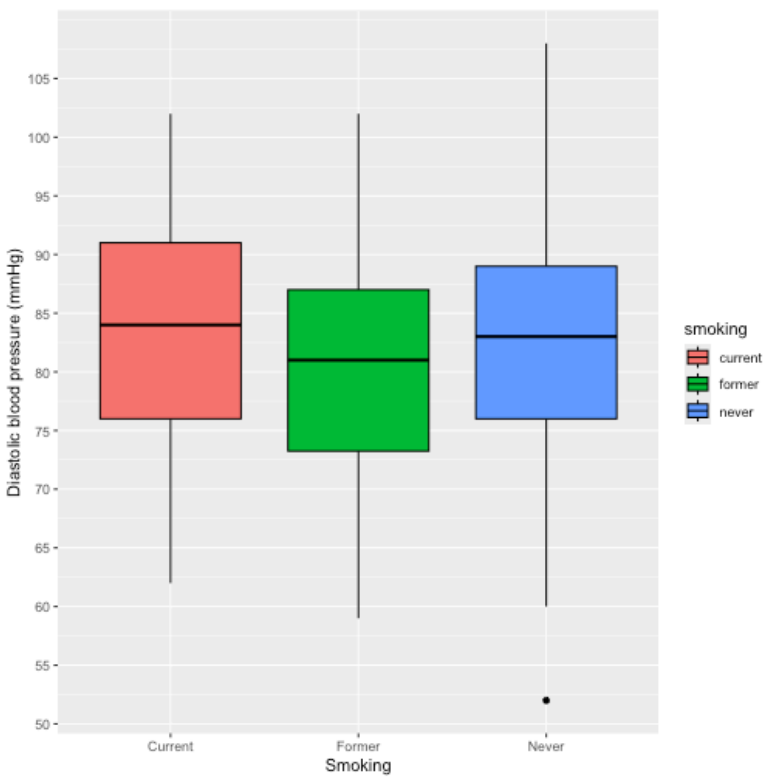
3.4.1 Axis

With `scales_x/y_<type>` (if continuous, binned, manual or discrete) we can control the details of the axis, such as breaks, labels, limits, mathematical transformation and so on.

For example, we can change the breaks on both x and y axis of diastolic_bp_boxplot:

- on y axis, the distance between breaks is now 10, we change it to 5
- on x axis, we change the labels of the axis with the first letter uppercase

```
diastolic_bp_boxplot <- diastolic_bp_boxplot +
  labs(x = "Smoking", y = "Diastolic blood pressure (mmHg)") +
  scale_y_continuous(breaks = seq(0, max(df$diastolic_bp_mmhg), 5)) +
  scale_x_discrete(labels = c("Current", "Former", "Never"))
diastolic_bp_boxplot
```



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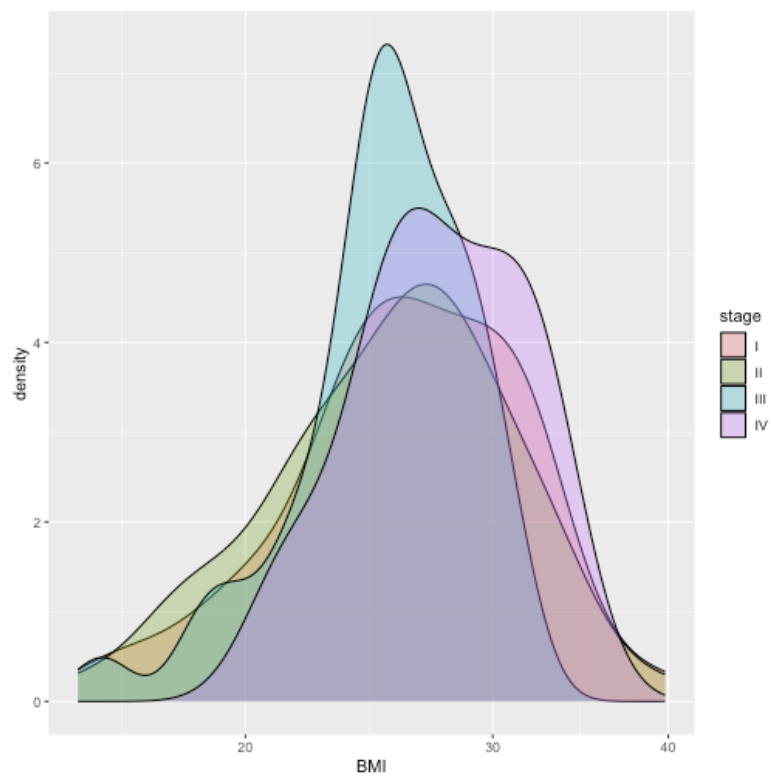
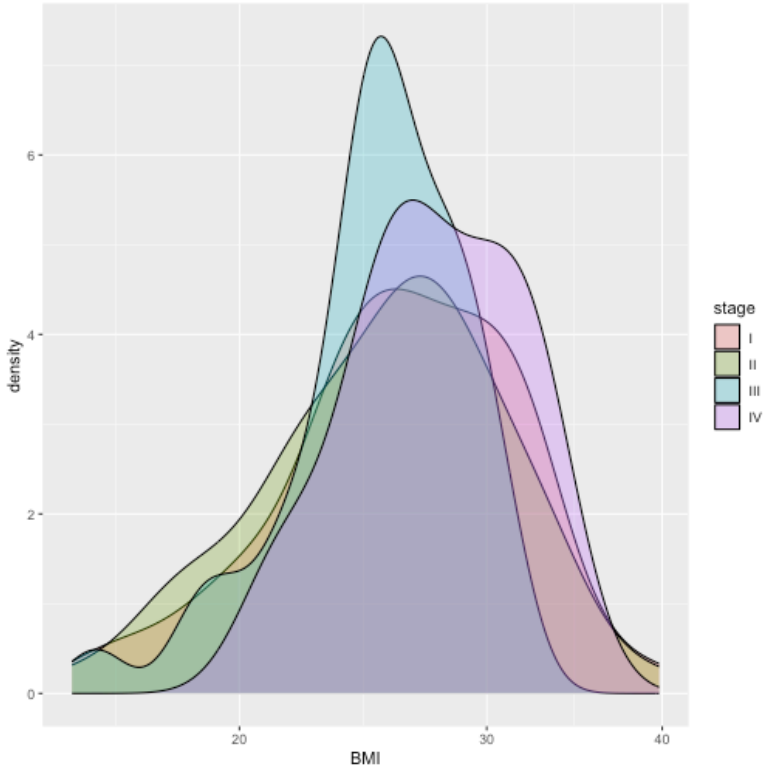
We can also apply mathematical transformation to continuous axis (like log scales):

```
BMI_stages_density +
  scale_x_continuous(trans = "log10")
```




```
BMI_stages_density +
  scale_x_log10()
```

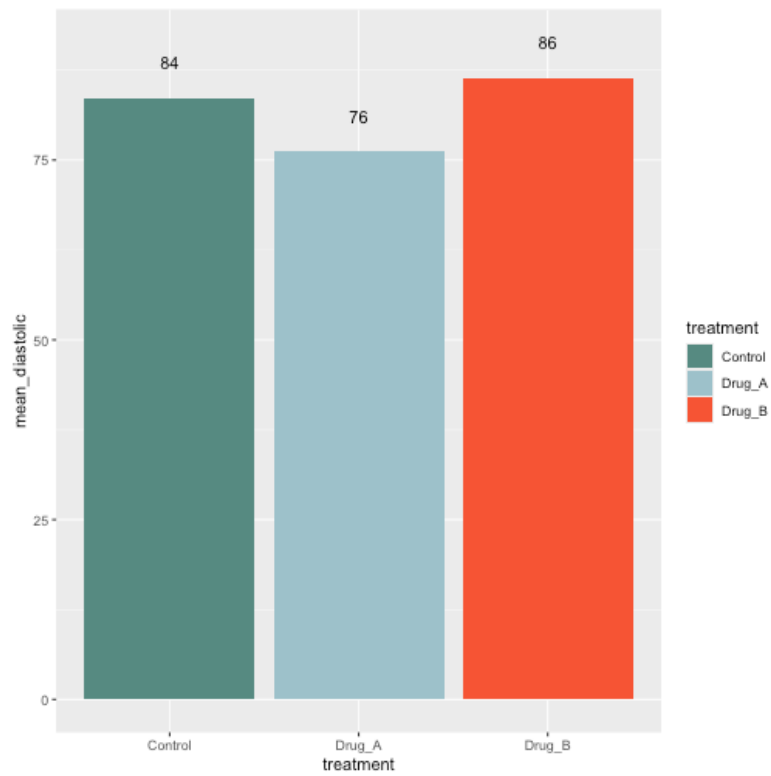


3.4.2 Colors

When we set `fill` or `color` aesthetics, we actually do not choose the colors that ggplot apply, we just say to it to make them different. With scales, we can set the colors we want. This site is very useful to choose colors.

For example, in stageI_female_diastolic_means_barplot, we want the Control to be gray, Drug_A to be blueish and Drug_B to be red. We can set it up like this:

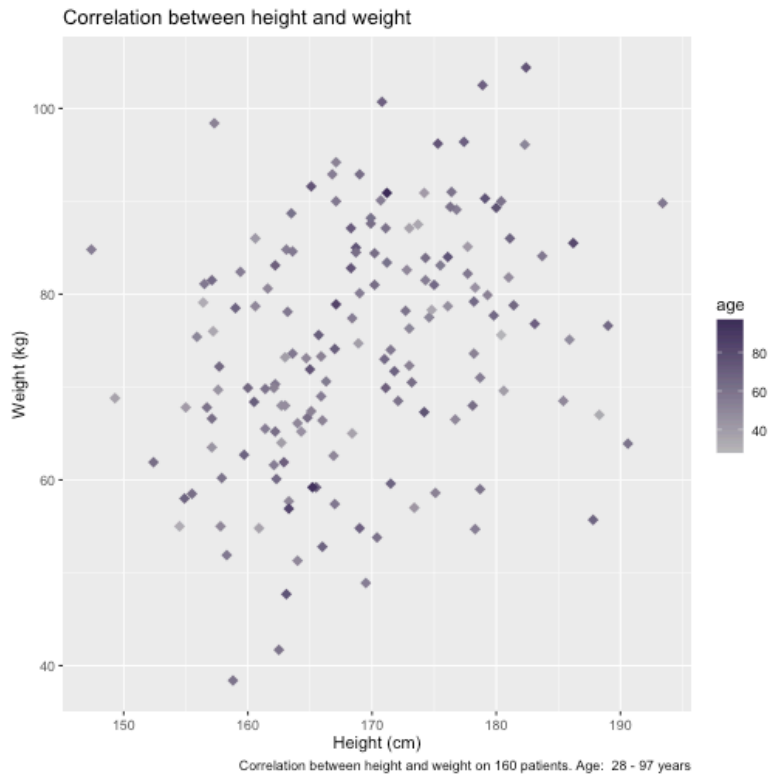
```
stageI_female_diastolic_means_barplot <- stageI_female_diastolic_means_barplot +
  scale_fill_manual(values = c("Control" = "#568A81",
                               "Drug_A" = "#9FC2CC",
                               "Drug_B" = "#F95738"))
stageI_female_diastolic_means_barplot
```



See this insite.) on hex color codes.

We can also change the color bar of corr_plot:

```
corr_plot <- corr_plot +
  scale_color_gradient(low = "#B8B7BA", high = "#3E2F5B")
corr_plot
```



See this stackoverflow post for some useful tips and explanation on color scales in ggplot.

Exercise 3.1. Create a boxplot of platelets_10e9l in males, divided by stage (x axis) and treatment (fill). Then:

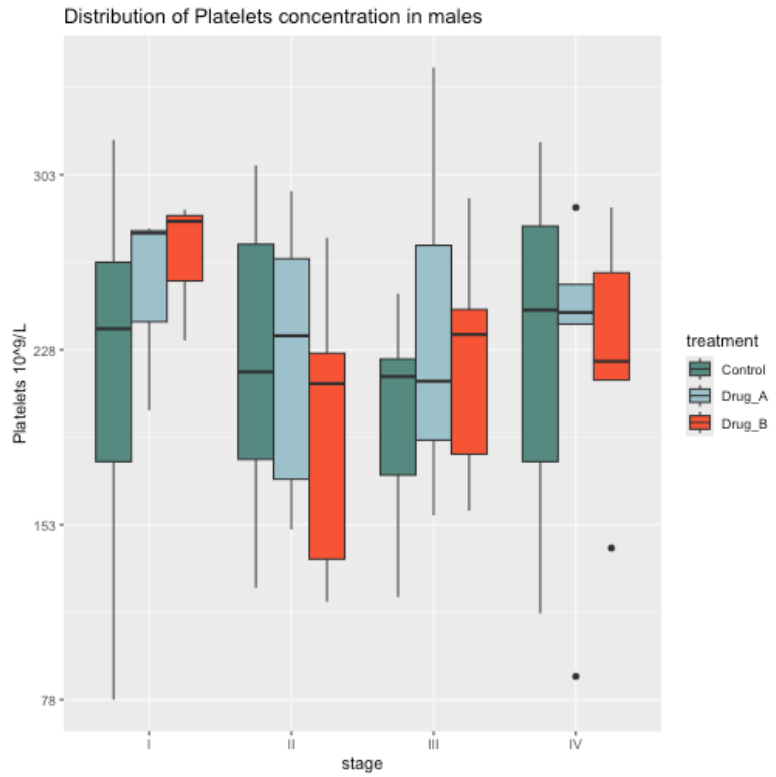
- y axis label should be "Platelets 10^9/L", with breaks every 75 starting from the minum value of platelets_10e9l up to the maximum value of platelets_10e9l
- Treatment colors should be the same as the ones in stageI_female_diastolic_means_barplot
- The title of the plot shoul be meaningful

Solution

```
males_df <- df |>
    filter(sex == "Male")

ggplot(males_df) +
    geom_boxplot(aes(x = stage, y = platelets_10e9l, fill = treatment), position = position_dodge
    labs(y = "Platelets 10^9/L",
```

```
        title = "Distribution of Platelets concentration in males") +
    scale_fill_manual(values = c("Control" = "#568A81",
                                 "Drug_A" = "#9FC2CC",
                                 "Drug_B" = "#F95738")) +
    scale_y_continuous(breaks = seq(min(males_df$platelets_10e9l), max(males_df$platel
```



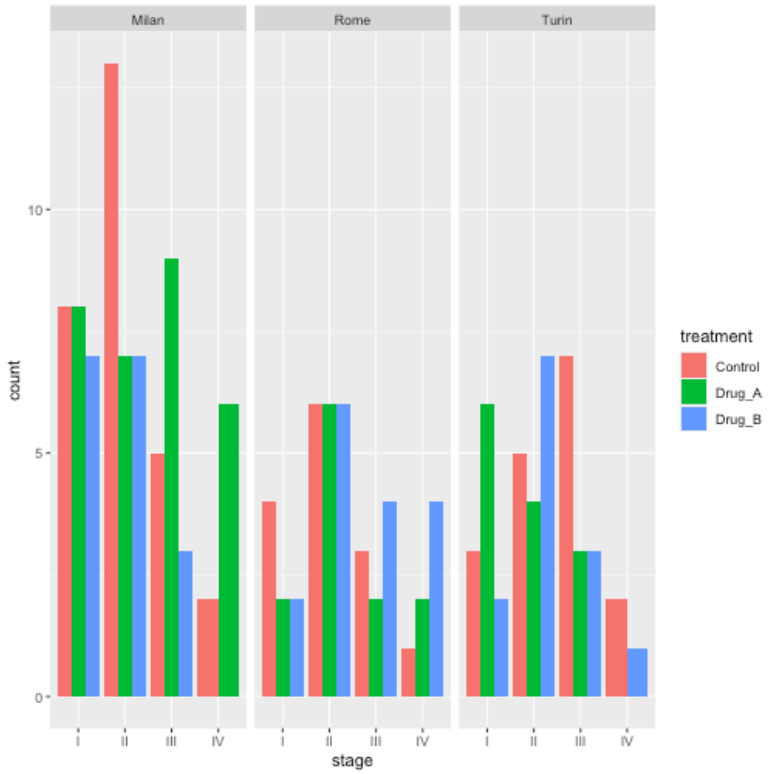
3.5 Facet

Faceting allow us to create a grid of panels, splitting data based on categorical variables.

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>() +
  facet_<type>() +
```

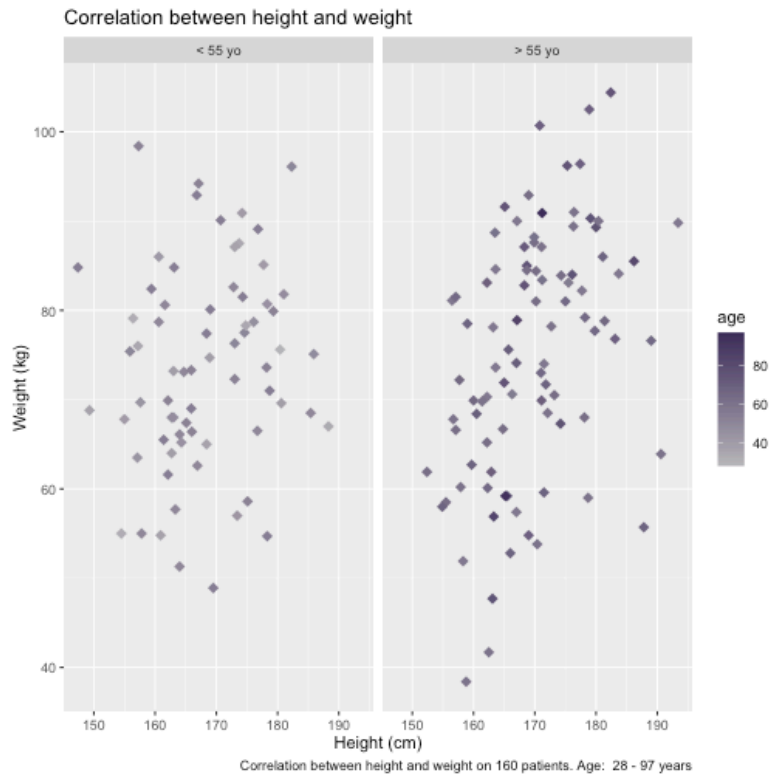
Let's see how we can rearrange the same data as in all_n_barplot but using multiple panels:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment), position = "dodge") +
    facet_wrap(~ site)
```



We can also customize the labels, similar to using scales:

```
corr_plot +
  facet_wrap(~ age > 55, labeller = labeller(`age > 55` = c("TRUE" = "> 55 yo", "FALSE" = "< 55 y
```



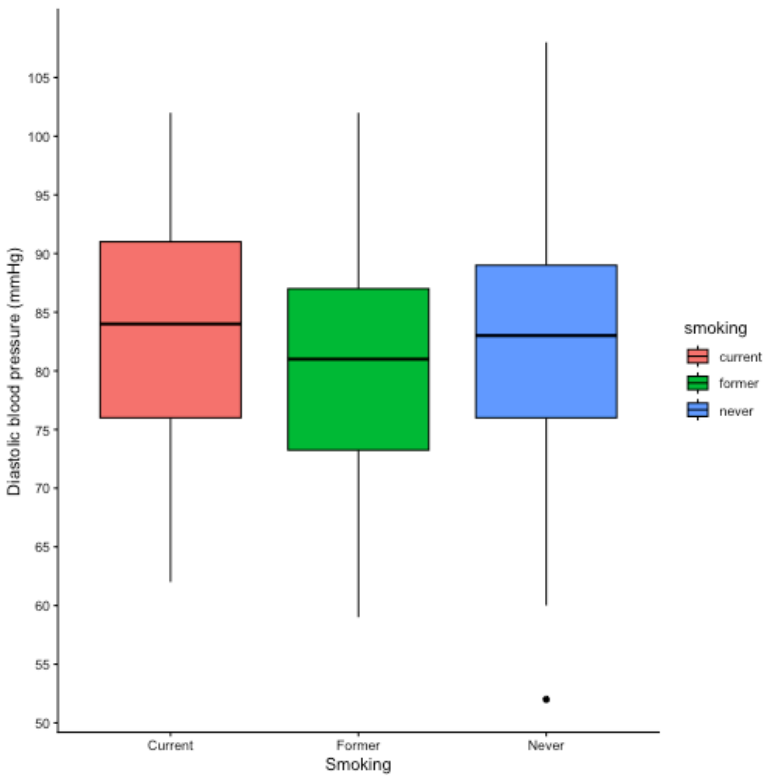
3.6 Theme

Ok, let's now see the last part (and maybe the most useful) of code. Theme allow us to control practically every single elements of the plot (except for the data): lines, font, labels, breaks, etc.

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>() +
  facet_<type>() +
  theme_<type>
```

There are few pre-installed theme, and we can further customize things with `theme`. Let's firt apply classic theme to diastolic_bp_boxplot:

```
diastolic_bp_boxplot <- diastolic_bp_boxplot +
  theme_classic()
diastolic_bp_boxplot
```

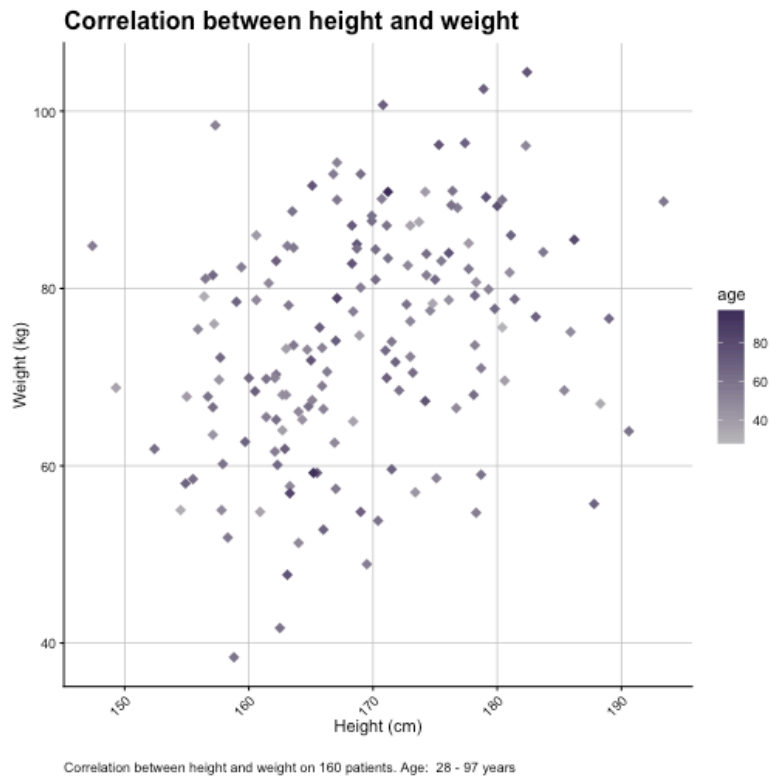


And now let's customize corr_plot:

- Title in bold and 16 pt size
- Add major grid lines in gray and 0.3 pt size
- Rotate x axis labels to 45°
- Change the adjustment of plot caption to left
- Add bottom margin to x title

```
corr_plot <- corr_plot +
  theme_classic() +
  theme(plot.title = element_text(face = "bold", size = 16),
        panel.grid.major = element_line(colour = "gray", linewidth = 0.3),
        axis.text.x = element_text(angle = 45, hjust = 1),
        plot.caption = element_text(hjust = 0),
```

```
        axis.title.x = element_text(margin = margin(b = 12)))
corr_plot
```



3.7 Save the plots

To save a plot to a file, we can use the function `ggsave` like this:

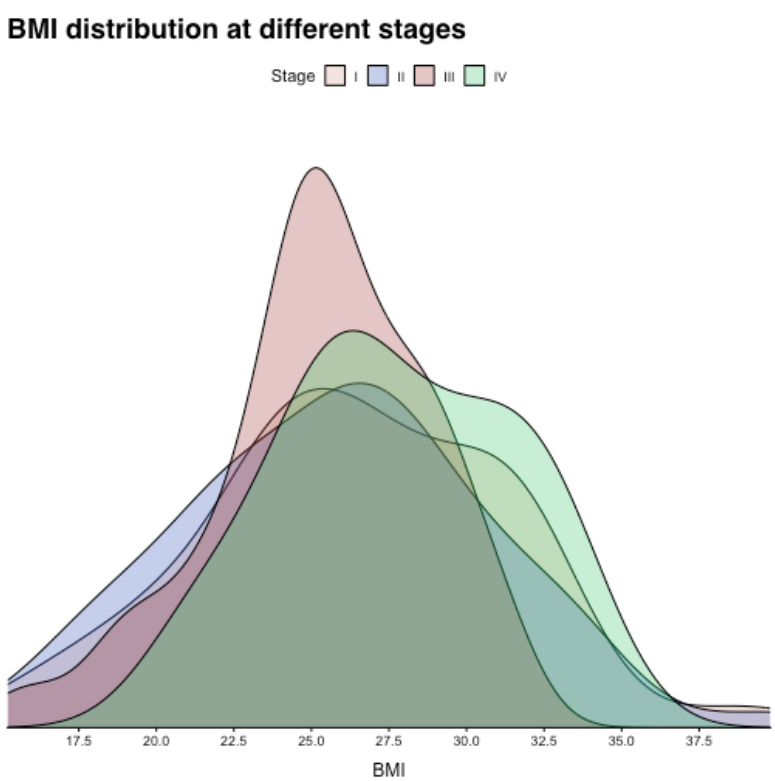
```
ggsave(plot = corr_plot, filename = "output/height_weight_corr-plot.pdf", units = "mm"
```

We can set units, width and height. See this article for further details about dimensions, dpi, dots etc.

3.8 Advanced examples

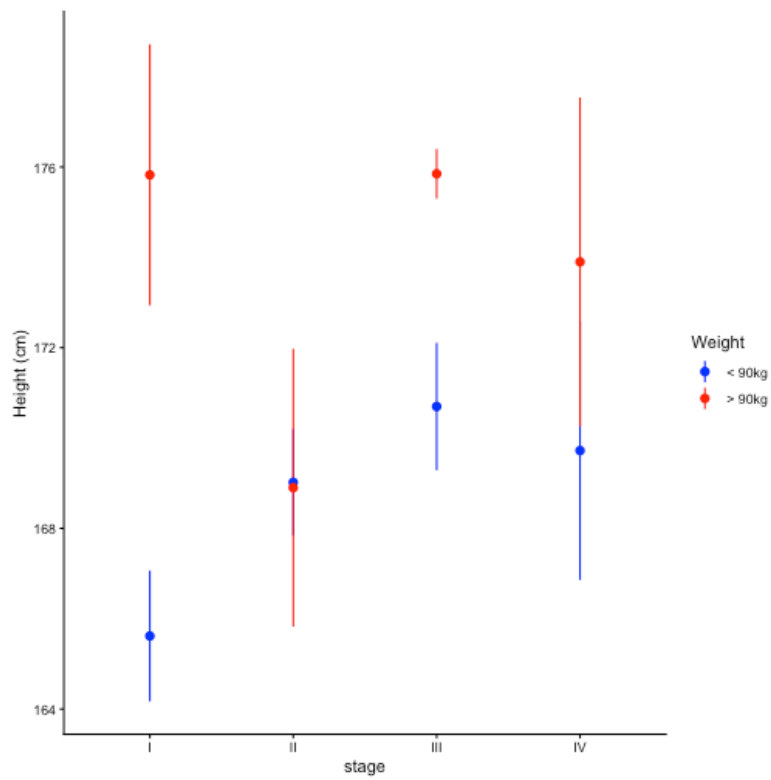
Here are some more advanced examples:

```
BMI_stages_density +
  labs(title = "BMI distribution at different stages", fill = "Stage") + # change labels
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) + # remove space at the bottom of y as
  scale_x_continuous(expand = expansion(mult = c(0, 0)), breaks = seq(0, max(df$BMI), 2.5)) + # r
  scale_fill_manual(values = c("I" = "#E0A890", "II" = "#2B59C3", "III" = "#9E2B25", "IV" = "#23C
  theme_classic() +
  theme(axis.line.y = element_blank(), # remove y-axes line
        axis.text.y = element_blank(), # remove y-axes labels
        axis.ticks.y = element_blank(), # remove y-axes ticks
        axis.title.y = element_blank(), # remove y-axes title
        legend.position = "top", # change legend position
        plot.title = element_text(size = 18, family = "sans", face = "bold", margin = margin(b =
        axis.title.x = element_text(size = 12, margin = margin(t = 10)),
        legend.key.size = unit(x = 5, units = "mm") # change legens key size
        )
```




```
ggplot(df) +
  stat_summary(aes(x = stage, y = height_cm, color = weight_kg > 90), fun.data = "mean_se", geom
  labs(y = "Height (cm)", color = "Weight") +
```

```
  scale_color_manual(labels = c("TRUE" = "> 90kg", "FALSE" = "< 90kg"), values = c("TRU
  theme_classic()
```



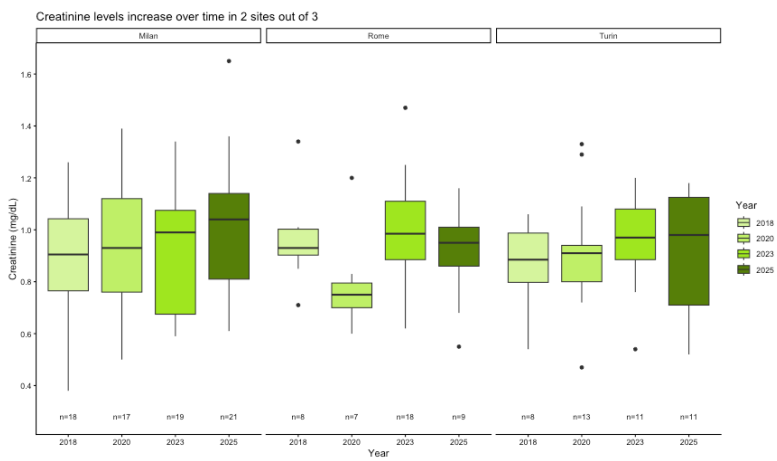
Exercise 3.2. Create a plot that shows if the levels of creatinine_mgdl changes depending on the year in each site. Set a proper scale for y axis, split the plot in panels, change the colors of the years, add a title. **Tip**: visit_year now is numeric, while we want it to be……..

bonus add the information about the numerosity in each group (year/site) at the bottom of x axis.

Solution

```
df$visit_year <- factor(df$visit_year)

df_n <- df |>
    group_by(site, visit_year) |>
    summarize(n = n(), .groups = "drop")

ggplot(df) +
```

```
    geom_boxplot(aes(x = visit_year, y = creatinine_mgdl, fill = visit_year),
                 position = position_dodge(width = 0.75)) +
    geom_text(data = df_n,
              aes(x = visit_year, y = min(df$creatinine_mgdl) - 0.1, label = paste0("n=", n)),
              size = 3) +
    labs(y = "Creatinine (mg/dL)",
         title = "Creatinine levels increase over time in 2 sites out of 3",
         x = "Year",
         fill = "Year") +
    scale_fill_manual(values = c("2018" = "#D6F49D",
                                 "2020" = "#C1EF6C",
                                 "2023" = "#A2E723",
                                 "2025" = "#59810E")) +
    scale_y_continuous(breaks = seq(0, max(df$creatinine_mgdl), 0.2)) +
    facet_wrap(~ site) +
    theme_classic()
```